

July 24, 2026 02:28 AM GMT

Greater China Semiconductors | Asia Pacific

TSMC preview and CoWoS update – CPU, GPU, ASIC, Optical, China AI Compute

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific

Industry View

Attractive

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Strong AI Semi Outlook

- Top ideas:
 - **OW:**
 - **AI:** MediaTek (Top Pick), TSMC, SMIC, Aspeed, Alchip, KYEC, ASE, FOCI, ASMPT, AllRing, GUC
 - **Memory:** Macronix (Top Pick), AP Memory, Nanya Tech, Winbond, GigaDevice
 - **China – AI/Semis/WFE:** Iluvatar, Cambricon, NAURA Tech, AMEC, USI
 - **Testing Equipment and Consumables:** WinWay, MPI, Hon Hai Precision, Gudeng
 - **Mature nodes:** UMC
 - **EW:** OmniVision, Phison, MetaX, Realtek, GlobalWafers
 - **UW:** WIN Semi, Silergy, ASMedia

Long-term demand drivers:

- **Chip inflation:** We expect “price elasticity” to affect demand for tech products. Rising wafer, OSAT, and memory costs create more margin headwinds for chip designers in 2026.
- **AI cannibalization:** Besides demand weakness (AI to replace some human jobs), we see the semi supply chain also prioritizing AI semis over non-AI semis, e.g., T-Glass and memory shortage.
- **China – AI: DeepSeek is triggering inferencing AI demand:** DeepSeek has demonstrated cheaper inferencing. Local foundry supply chain is also becoming more capable for AI GPU production.

Valuation Comparison: Foundry, Back-end, Memory, IDM and Semi Cap

							Dividend	FCF	Market	3M Avg. Daily	P/E Ratio (x)			EPS Growth			ROAE			P/B Ratio (x)		
Ticker	Company	Price 7/21/2026	Curr ency	Price Target	Upside/ Downside	Rating	Yield (%)	Yield (%)	Cap (US\$M)	Trading (US\$M)	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e
Foundry																						
2330.TW	TSMC	2,410.0	TWD	2,988.0	24%	O	1.4%	1.8%	1,973,075	3,041.4	21.7	16.6	13.1	67%	31%	27%	44%	41%	38%	8.1	5.8	4.3
2303.TW	UMC	134.5	TWD	138.0	3%	O	5.1%	5.3%	53,616	1,035.8	29.9	23.7	19.5	35%	26%	22%	14%	17%	19%	4.1	3.8	3.6
0981.HK	SMIC	75.3	HKD	85.0	13%	O	0.0%	-5.0%	76,261	1,339.8	NM	NM	NM	142%	45%	20%	7%	10%	11%	NM	NM	NM
5347.TWO	Vanguard	155.5	TWD	180.0	16%	E	4.9%	3.1%	9,125	199.7	28.5	23.0	18.7	31%	24%	23%	12%	14%	17%	3.4	3.2	3.0
1347.HK	Hua Hong	168.4	HKD	118.0	-30%	E	0.1%	-4.6%	37,418	784.1	NM	NM	NM	584%	55%	27%	5%	7%	8%	NM	NM	NM
6770.TW	PSMC	65.1	TWD	111.0	71%	O	0.0%	5.6%	8,683	471.2	NM	NM	NM	-142%	367%	29%	4%	11%	12%	1.4	1.1	0.9
3105.TWO	WIN Semi	339.5	TWD	300.0	-12%	U	0.5%	-3.8%	2,264	408.5	NM	38.2	26.4	59%	40%	45%	6%	9%	12%	3.4	3.3	3.2
8086.TWO	AWSC	119.5	TWD	130.0	9%	U	1.1%	-3.9%	742	30.2	18.9	16.7	13.5	89%	13%	24%	14%	14%	16%	2.5	2.3	2.0
Mean:							1.6%	-0.2%			23.7	26.0	19.5	108%	75%	27%	13%	15%	16%	3.8	3.2	2.8
Median:							0.8%	-1.0%			25.1	23.0	18.7	63%	35%	25%	10%	13%	14%	3.4	3.2	3.1
Memory																						
603986.SS	Giga Device	475.5	CNY	888.0	87%	O	0.3%	4.1%	44,120	3,895.0	28.5	15.7	13.5	574%	81%	16%	47%	54%	43%	10.4	6.6	4.8
2408.TW	Nanya Tech	405.0	TWD	580.0	43%	O	0.0%	13.6%	39,415	1,374.6	6.5	3.6	5.4	NM	NM	NM	77%	68%	29%	3.3	1.6	1.2
2344.TW	Winbond	155.5	TWD	288.0	85%	O	0.0%	16.6%	22,096	1,026.7	7.3	4.1	3.6	NM	NM	NM	61%	59%	41%	3.4	1.9	1.2
8299.TWO	Phison	1,855.0	TWD	2,588.0	40%	E	1.5%	1.6%	12,767	515.8	5.4	14.0	33.2	758%	-61%	-58%	79%	23%	10%	3.1	3.3	3.4
SIMO.O	Silicon Motion	252.6	USD	400.0	58%	O	2.2%	2.1%	8,487	277.1	26.8	14.4	9.2	198%	86%	57%	33%	45%	49%	7.7	5.5	3.8
2337.TW	Macronix	123.0	TWD	220.0	79%	O	4.6%	17.2%	7,206	624.1	NM	NM	NM	-652%	154%	3%	36%	57%	37%	3.7	2.1	1.4
Mean:							1.4%	9.2%			14.9	10.4	13.0	219%	65%	5%	55%	51%	35%	5.3	3.5	2.6
Median:							0.9%	8.8%			7.3	14.0	9.2	386%	84%	10%	54%	55%	39%	3.6	2.7	2.4
Backend																						
3711.TW	ASEH	633.0	TWD	558.0	-12%	O	2.2%	6.0%	87,212	488.2	37.6	23.5	15.5	83%	60%	52%	19%	26%	33%	6.5	5.5	4.6
600584.SS	JCET	84.7	CNY	50.0	-41%	E	0.3%	5.7%	21,054	2,310.3	58.6	42.4	35.5	65%	38%	19%	8%	11%	11%	4.7	4.3	3.9
601231.SS	USI	25.3	CNY	45.4	80%	O	1.4%	-7.4%	8,008	271.8	21.3	14.4	12.3	42%	48%	17%	13%	17%	17%	2.6	2.3	2.0
2449.TW	KYEC	279.0	TWD	388.0	39%	O	1.8%	1.6%	10,772	375.4	28.0	18.5	14.0	11%	51%	33%	23%	29%	32%	6.0	4.9	4.0
3363.TWO	FOCI	523.0	TWD	708.0	35%	O	0.2%	4.8%	1,737	73.9	NM	NM	NM	NM	NM	NM	-2%	55%	79%	NM	NM	NM
Mean:							1.4%	1.5%			24.7	16.4	13.1	50%	49%	30%	16%	21%	23%	4.9	4.3	3.6
Median:							1.6%	3.7%			24.7	16.4	13.1	54%	50%	26%	16%	21%	25%	5.3	4.6	3.9
Power Semis/ IDM																						
688396.SS	CR Micro	65.3	CNY	51.6	-21%	U	0.2%	1.6%	11,899	411.8	54.8	36.7	26.5	138%	49%	39%	6%	9%	11%	3.2	3.0	2.7
600460.SS	Silan Micro	32.9	CNY	26.9	-18%	U	0.5%	2.1%	7,599	598.6	39.9	26.8	20.9	244%	49%	28%	10%	14%	15%	3.9	3.5	3.0
603290.SS	StarPower	88.8	CNY	120.0	35%	E	1.0%	0.0%	2,953	220.1	56.7	33.3	22.2	-7%	70%	50%	5%	8%	12%	2.9	2.7	2.5
300373.SZ	Yangjie	98.5	CNY	136.0	38%	O	1.4%	-2.8%	7,389	431.7	33.6	24.0	17.9	27%	40%	34%	15%	18%	21%	4.7	4.1	3.4
Mean:							0.8%	0.2%			46.2	30.2	21.9	100%	52%	38%	9%	12%	15%	3.7	3.3	2.9
Median:							0.7%	0.8%			47.3	30.0	21.6	82%	49%	36%	8%	11%	14%	3.6	3.2	2.9
Semicap / Equipment/ Materials/ Supply Chain																						
002371.SZ	NAURA	744.6	CNY	818.0	10%	O	0.6%	-0.9%	74,956	1,376.4	58.6	40.0	31.5	45%	46%	27%	19%	24%	25%	10.5	8.7	7.2
688012.SS	AMEC	407.9	CNY	550.0	35%	O	0.0%	17.6%	35,591	1,524.8	66.8	48.9	35.7	90%	37%	37%	16%	18%	20%	9.6	8.0	6.5
7769.TW	Hon Precision	6,500.0	TWD	10,008.0	54%	O	2.5%	NA	36,925	193.3	50.3	29.2	17.5	69%	72%	67%	37%	53%	65%	NM	13.6	9.8
6223.TWO	MPI	5,875.0	TWD	7,500.0	28%	O	0.0%	-5.4%	18,177	220.3	85.7	41.0	20.4	109%	109%	101%	38%	54%	69%	NM	NM	NM
6515.TW	Winway	6,655.0	TWD	15,000.0	125%	O	0.9%	1.3%	7,513	116.6	NM	32.4	15.2	104%	116%	113%	45%	67%	84%	NM	NM	NM
6488.TWO	GlobalWafers	1,205.0	TWD	750.0	-38%	E	2.7%	1.7%	18,192	222.2	53.9	36.1	28.4	46%	50%	27%	11%	15%	18%	5.8	5.3	4.8
0522.HK	ASM Pacific	170.7	HKD	248.0	45%	O	1.8%	6.5%	8,679	110.4	43.3	25.3	18.1	82%	72%	39%	9%	15%	19%	3.8	3.4	3.1
688234.SS	SICC	101.1	CNY	112.1	11%	O	0.6%	NA	6,808	384.4	NM	NM	NM	NM	NM	NM	1%	3%	3%	6.8	6.6	6.4
ACMR.O	ACM Research	82.5	USD	130.0	58%	O	1.1%	-0.7%	5,414	134.0	28.4	18.3	14.4	109%	56%	27%	10%	14%	16%	2.6	2.3	2.0
3680.TWO	Gudeng Precision	481.0	TWD	648.0	35%	O	0.0%	-6.1%	1,459	48.9	24.7	17.5	12.7	106%	42%	38%	15%	19%	22%	3.5	3.0	2.5
6187.TWO	AlifRing Tech	998.0	TWD	1,580.0	58%	O	2.8%	1.1%	3,033	100.2	39.2	25.1	22.2	64%	56%	13%	31%	39%	35%	NM	NM	NM
Mean:							1.2%	1.7%			50.1	31.4	21.6	82%	66%	49%	21%	29%	34%	6.1	6.4	5.3
Median:							0.9%	1.1%			50.3	30.8	19.3	86%	56%	38%	16%	19%	22%	5.8	5.9	5.6

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Data as of July 21, 2026.

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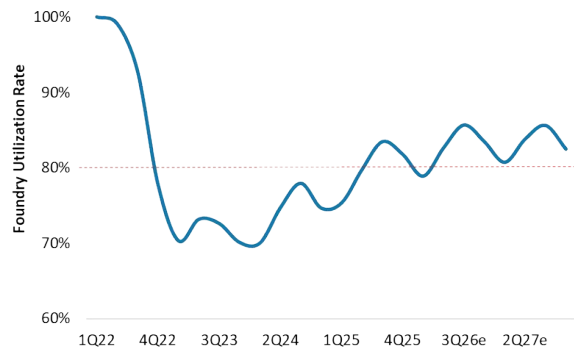
Valuation Comparison: Fabless, Power Semis, FPGA and Analog

										P/E Ratio (x)			EPS Growth			ROAE			P/B Ratio (x)			
		Price	Curr	Price	Upside/		Yield	FCF	Market	3M Avg. Daily												
Ticker	Company	7/21/2026	ency	Target	Downside	Rating	(%)	Yield (%)	Cap (US\$M)	Trading (US\$M)	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e
Fabless																						
AI accelerators																						
688256.SS	Cambricon	1,350.0	RMB	1,528.0	13%	O	0.0%	-2.2%	117,831	2,780.7	106.8	69.4	50.8	160%	54%	37%	48%	51%	46%	45.4	28.8	19.4
9903.HK	Iluvatar	576.5	HKD	688.0	19%	O	NA	NA	18,749	123.1	NM	89.5	50.2	-83%	-805%	78%	-6%	26%	34%	26.8	20.7	14.6
688802.SS	MetaX	896.0	RMB	758.0	-15%	E	0.0%	NA	49,803	308.0	NM	645.9	359.5	-103%	2214%	80%	0%	4%	7%	27.2	26.1	24.3
Smartphone/Wireless Semis																						
2454.TW	MediaTek	3,670.0	TWD	5,588.0	52%	O	17.3%	0.9%	181,383	1,573.4	52.5	25.1	11.9	12%	109%	111%	25%	40%	55%	12.0	8.2	5.3
603501.SS	OmniVision	102.4	CNY	111.0	8%	E	0.5%	2.1%	17,158	498.7	30.8	21.2	17.2	0%	45%	23%	14%	18%	19%	3.9	3.4	2.9
300782.SZ	Maxscend	75.7	CNY	80.00	6%	U	0.1%	-2.2%	5,617	433.0	NM	83.0	44.8	-20%	-309%	85%	-2%	5%	9%	4.2	4.0	3.7
603160.SS	Goodix	51.5	CNY	58.0	13%	U	0.7%	-1.4%	3,292	71.0	43.1	38.1	34.5	-35%	13%	11%	6%	6%	7%	2.5	2.3	2.2
6462.TWO	Egis	99.9	TWD	105.0	5%	E	1.0%	-16.3%	288	9.6	NM	18.6	NM	-88%	-462%	NA	-1%	5%	NA	1.0	1.0	NA
IP/Design Service																						
3661.TW	Alchip	3,490.0	TWD	5,088.0	46%	O	1.0%	5.1%	8,985	303.9	23.0	19.3	16.0	124%	19%	21%	28%	28%	30%	NM	NM	NM
301269.SZ	Empyrean Technology	99.9	CNY	95.0	-5%	E	0.1%	NA	7,569	283.4	NM	NM	NM	283%	136%	70%	4%	10%	15%	NM	NM	NM
3443.TW	GUC	4,115.0	TWD	5,688.0	38%	O	0.8%	4.7%	17,413	339.1	93.4	33.1	21.7	58%	182%	52%	40%	71%	66%	NM	NM	NM
6531.TW	AP Memory	858.0	TWD	1,555.0	81%	O	1.2%	4.1%	4,384	181.2	48.6	25.0	14.1	128%	95%	77%	22%	35%	47%	9.7	7.7	5.8
PC/Cloud/Display Semis																						
3034.TW	Novatek	476.0	TWD	414.0	-13%	U	6.1%	14.3%	9,146	99.2	13.8	13.5	15.8	29%	2%	-15%	30%	28%	23%	3.9	3.7	3.7
2379.TW	Realtek	753.0	TWD	717.0	-5%	E	4.8%	5.0%	12,131	114.6	17.6	17.7	17.5	48%	0%	1%	38%	34%	32%	6.2	5.8	5.5
5269.TW	ASMedia	1,410.0	TWD	1,482.0	5%	U	4.1%	0.9%	3,324	80.4	15.4	15.8	14.9	25%	-3%	6%	18%	16%	15%	2.6	2.3	2.1
5274.TWO	Aspeed	14,100.0	TWD	21,323.6	51%	O	NA	1.3%	16,831	151.9	NM	NM	NM	NA	NA	NA	85%	79%	81%	NM	NM	NM
4966.TWO	Parade	583.0	TWD	818.0	40%	O	2.9%	1.6%	1,433	61.8	19.2	13.5	11.6	-11%	42%	17%	10%	14%	15%	2.0	1.8	1.6
2458.TW	Elan Micro	165.5	TWD	250.0	51%	O	4.8%	9.6%	1,496	27.6	15.8	13.4	13.3	24%	18%	1%	27%	28%	26%	4.1	3.6	3.3
HIMX.O	Himax	12.7	TWD	17.4	37%	E	2.7%	8.3%	2,164	69.4	31.9	17.5	8.7	65%	82%	102%	8%	14%	26%	2.6	2.4	2.1
Analog IC																						
6415.TW	Silergy	461.5	TWD	388.0	-16%	U	1.4%	4.8%	5,655	109.2	45.5	28.0	21.8	59%	62%	29%	10%	15%	17%	4.4	3.9	3.5
300661.SZ	SG Micro	110.1	CNY	82.0	-25%	E	0.3%	1.1%	9,481	441.4	80.7	56.5	45.3	49%	43%	25%	15%	18%	20%	NM	NM	NM
MCU																						
4919.TW	Nuvoton	141.0	TWD	98.0	-30%	U	0.8%	14.5%	1,869	101.1	NM	32.0	17.9	-69%	-457%	79%	-4%	14%	21%	4.7	4.1	3.3
688018.SS	Espressif	106.0	CNY	125.0	18%	O	NA	NA	3,429	165.9	NM	NM	NM	NA	NA	NA	12%	14%	15%	NM	NM	NM
FPGA																						
002049.SZ	Unigroup Guoxin	65.3	CNY	57.0	-13%	U	0.4%	1.5%	7,705	496.6	26.8	20.6	17.7	44%	30%	17%	14%	16%	16%	3.6	3.1	2.7
1385.HK	Shanghai Fudan	27.4	HKD	47.3	73%	O	0.1%	0.1%	6,905	30.2	21.3	17.5	15.6	353%	22%	12%	16%	16%	16%	3.1	2.7	2.3
IC distributor																						
3702.TW	WPG	111.5	TWD	160.0	43%	O	7.2%	-27.2%	5,912	49.2	8.7	8.9	8.5	115%	-2%	5%	24%	21%	20%	1.8	1.7	1.6
3036.TW	WT Micro	196.0	TWD	349.0	78%	O	4.4%	-21.4%	7,832	91.8	7.9	7.0	6.2	107%	13%	13%	25%	25%	24%	1.8	1.6	1.4

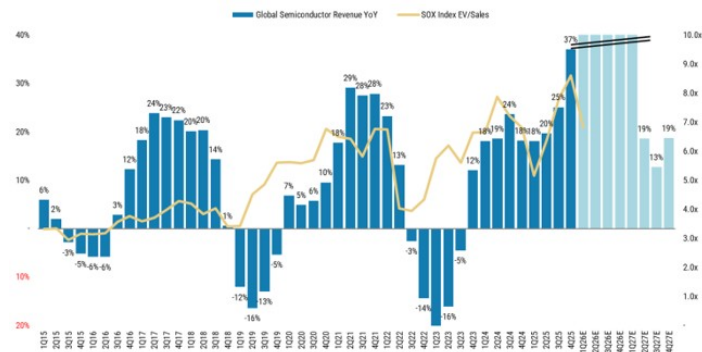
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates. Note: Data as of July 21, 2026.

Broader Semiconductor Cycle

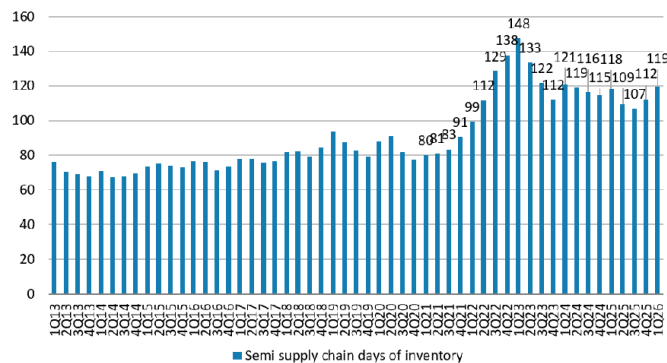
Logic semi foundry utilization could reach 80% in 2H26



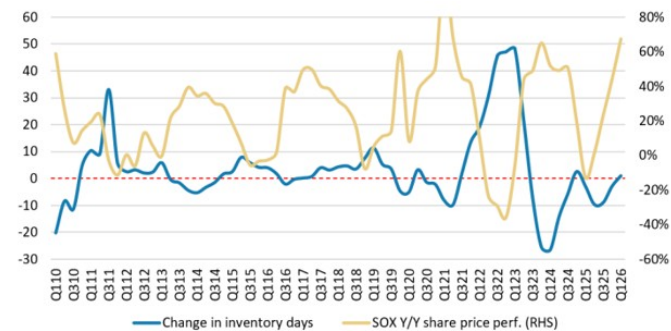
Excluding memory and NVIDIA's AI GPU revenue, non-AI semi growth is expected to decline in 2026



Semi supply chain – days of inventory as of 1Q26



Historically, when inventory days have gone down, the semi stock index has risen



Source: SIA, FactSet, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

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TSMC: 2Q26 Earnings Preview

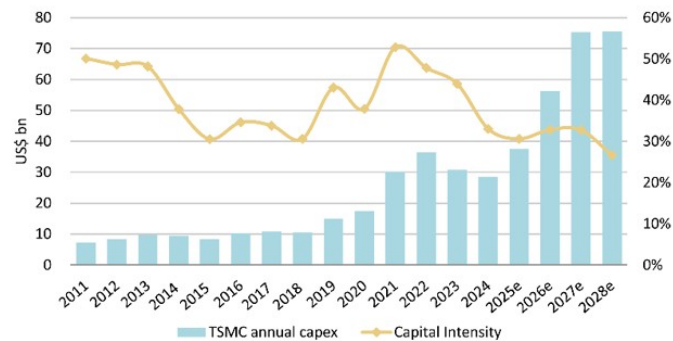
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TSMC: 2Q26 earnings preview and 3Q26 guidance

2Q26 earnings preview and 3Q26 guidance

(NT\$ bn)	2Q26				3Q26		
	Guidance	Preliminary	MS Est.	Consensus	Guidance	MS Est.	Consensus
Revenue	US\$39.0 - 40.2bn		1,267	1,265		1,436	1,382
Q/Q	+10% at mid-point in USD			11.5%			9.3%
Y/Y				35.4%			39.6%
GM (%)	65.5% - 67.5%		67.4%	67.1%		67.5%	66.2%
OpM (%)	56.0% - 58.0%		59.8%	58.5%		60.3%	57.8%
EPS (NT\$)			25.08	24.02		29.21	26.98

We saw TSMC 2026/2027e capex at US\$56bn and US\$75bn before 2Q26 earnings call



TSMC 3Q26/2026 guidance scenario

Details of the scenario	Scenario 1	Scenario 2	Scenario 3
PROBABILITIES	20%	60%	20%
3Q26 USD revenue growth	15% Q/Q	10-15% Q/Q	5-10% Q/Q
3Q26 gross margin	70%	67-68%	66-67%
2026 full-year revenue growth	Around 40%	Close to 40%	Around 35%
2026 full-year capex in USD	Raised to US \$60bn	Raised to US \$56bn	Unchanged at US \$54-56bn
Five-year AI semi revenue CAGR	Raised to 80%	Raised to 70%	Unchanged at mid-high 50s (%)
AI demand sustainability	Until the end of the decade	Strong until 2028	Could be affected by memory costs
Potential change in stock price (%)	Up 3-5%	Up 1-3%	Down 3-5%

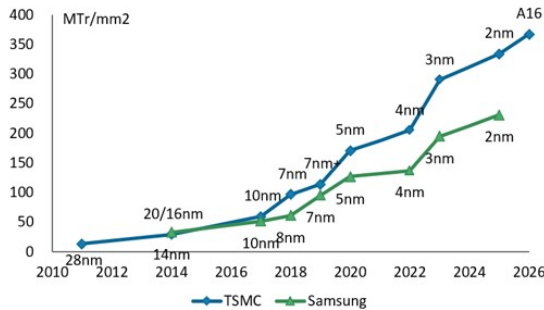
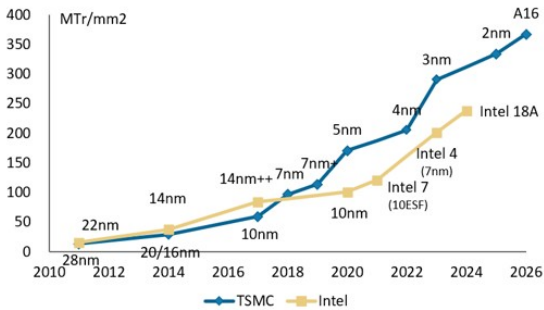
Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC remains the only game in town given technology leadership and tight EUV supply

TSMC still leads in technology roadmap as well as the logic density

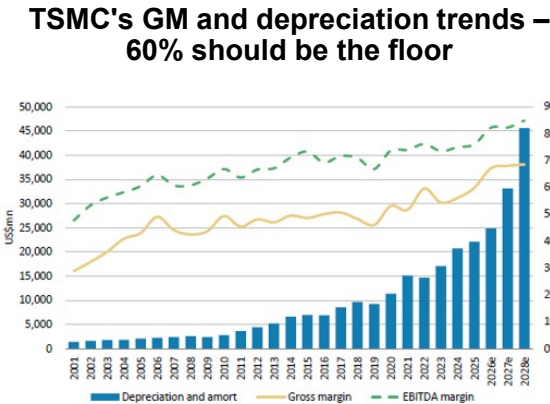
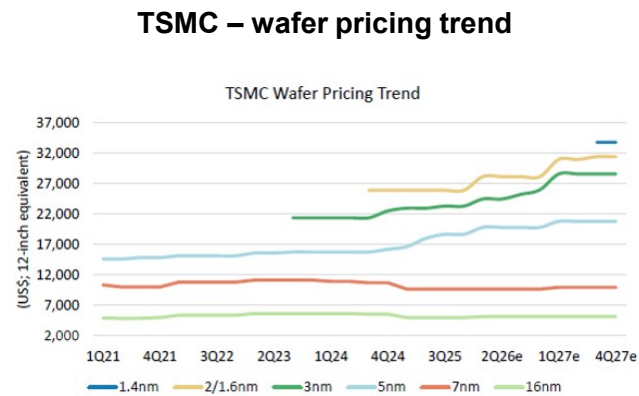
CY	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025e	2026e
Intel		22nm FinFET		14nm FinFET						10nm FinFET		Intel 17 FinFET	Intel 14 (EUV)	Intel 13	Intel 20A GAA	Intel 18A GAA
										10nm			7nm		5nm	
TSMC	28nm			20nm	16nm FinFET	16nm+ FinFET	10nm FinFET	7nm	7nm Pro	5nm	5nm Pro		3nm FinFET		2nm GAA	1.6nm GAA
							12nm	22nm	7nm+ (EUV)		6nm	4nm	N3e			
Samsung	32nm	28nm			14nm FinFET		10nm FinFET	8nm	7nm (EUV)	5nm		4nm	3nm GAA		2nm GAA	
						28FDS		28FDS-RF	14nm LPU	28FDS-eMRAM	18FDS					

Logic density comparison between TSMC, Intel, and Samsung foundry



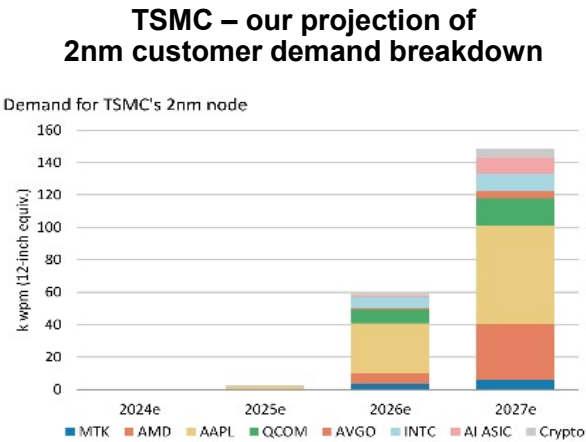
Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC should be able to hike its leading-edge prices 5-10% in 2027, reflecting its value to customers

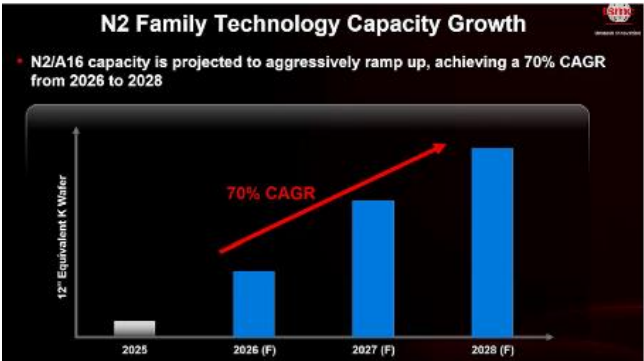


Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC 2nm demand remains very strong

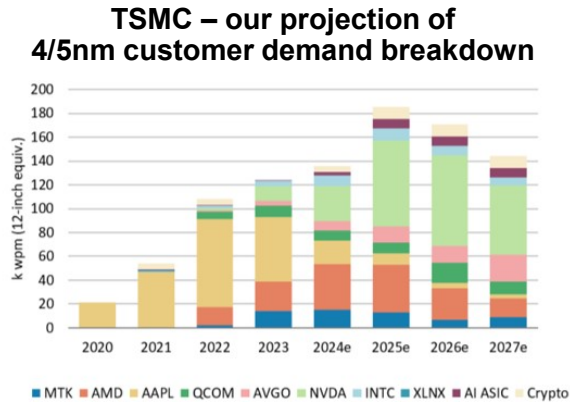
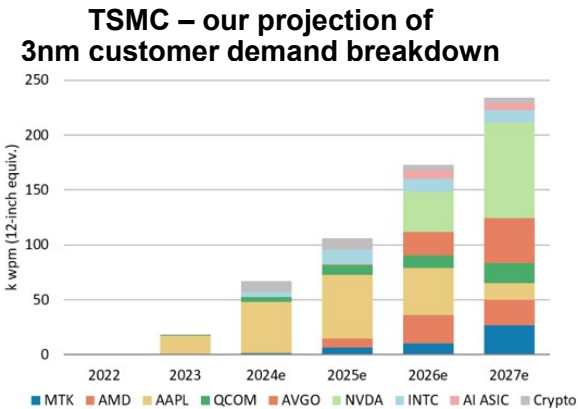


TSMC suggests N2 capacity to achieve 70% CAGR from 2026-2028e

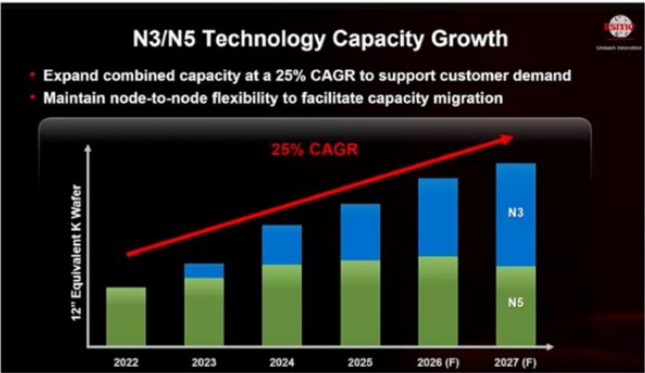


Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC N5 capacity to drop in 2027 while N3 should keep increasing, similar to our demand forecast



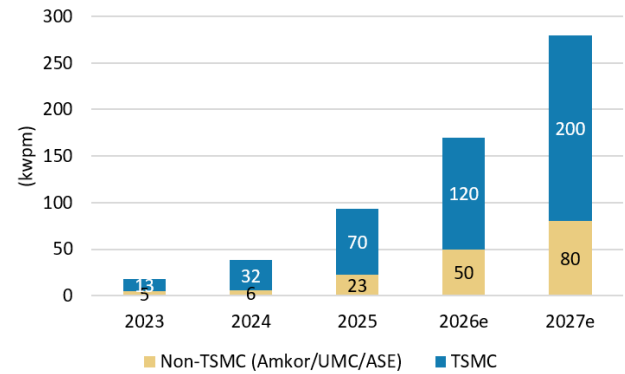
TSMC indicates that N5 capacity will drop in 2027 while N3 capacity should keep increasing, similar to our demand forecast



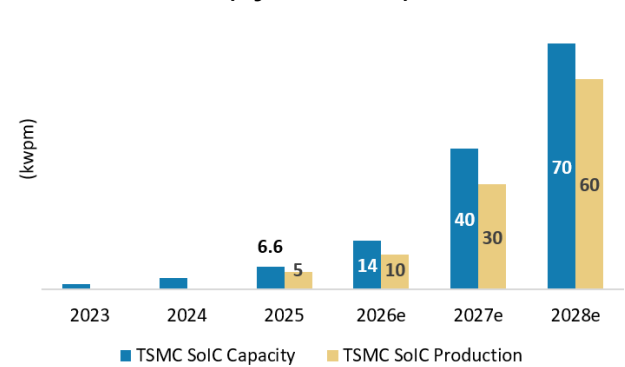
Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC’s advanced packaging capacity expansion should also be the key focus in 2026/27

Our projection of CoWoS Supply Capacity Breakdown (by Year End)



Our projection of TSMC SoIC Supply Capacity Breakdown (by Year End)



TSMC is also addressing strong advanced packaging capacity build



Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC's capex is tracking at US\$75bn in 2027

TSMC – fab roadmap

Node	Location	Fab	Capacity				
			2025	2026	2027	2028	2029
N3	Tainan	F18 - P4/P5/P6/P8/P9	110k	160-170k			
	Arizona	F21 - P2		20k			
	Japan	F23 - P2				15k	
Total			110k	180-190k	190k	200-205k	
N2/A16	HsinChu	F20 - P1/P2	20k	30k			
	Tainan	F22 - P7/P8/P9			40-50k	60k	
	KaoHsiung	F22 - P1/P2/P3/P4/P5/P6	25k	60k	80-90k	100k	
	Arizona	F21 - P3/P4/P5/P6				20k	
Total			45k	90-100k	150-170k	210k	
A14	HsinChu	F20 - P3/P4			10-20k	30-40k	40k
A14/A13/A12	Taichung	F25 - P1/P2/P3/P4			10-20k	30-40k	80k
Total					20-40k	60-80k	120k
A10	Tainan	F26 - P1/P2/P3/P4					5k
N40/28/22/12	Japan	F23 - P1	25k				
N28/16	Germany	F24			5k	40k	

TSMC – fab details

Front-end Plant	Location	Focused Technology
F18 - P4/P5/P6/P8/P9	Tainan	N3
F20 - P1/P2/P3/P4	HsinChu	N2/A14
F21 - P2	Arizona	N3
F22 - P1/P2/P3/P5/P6	KaoHsiung	N2
F22 - P7/P8/P9	Tainan	N2
F22 - P4	KaoHsiung	A16
F23 - P2	Kumamoto	N3
F25 - P1/P2/P3/P4	Taichung	A14
F26 - P1/P2/P3/P4	Tainan	A10

TSMC states that its Arizona Fab P2 tool move-in will be in 2H26, in line with our 20kwpw capacity addition for 3nm



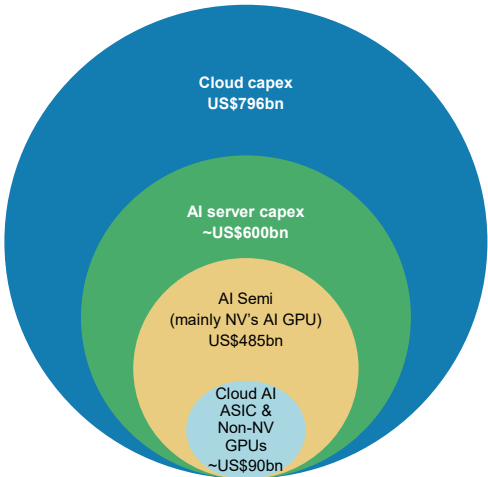
Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Morgan Stanley

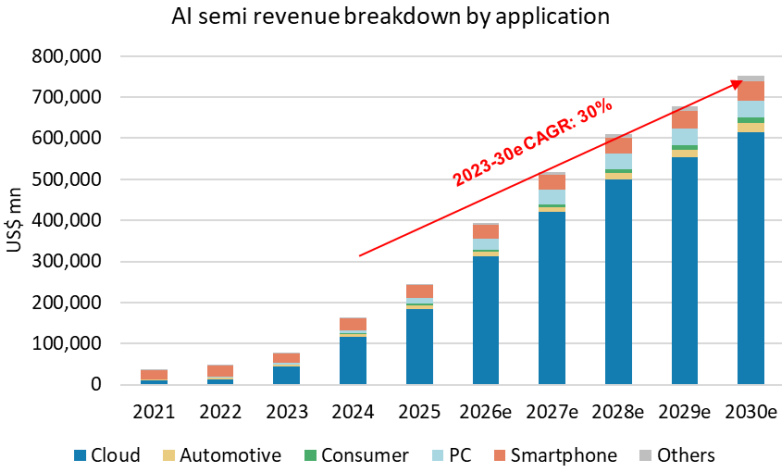
AI Semi Supply Chain and Niche Memory

Global Semi Industry Market Could Reach US\$1.5tn by 2030e, with AI Semis Contributing Half

Our supply chain data-driven bull case assumption is that cloud AI Semi TAM could grow to US\$485bn in 2026e



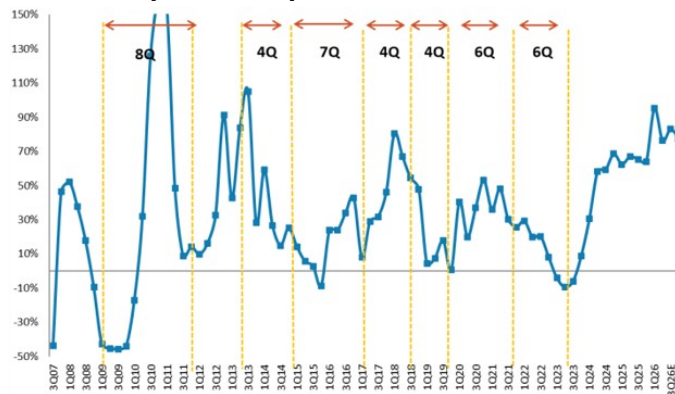
We expect AI semi TAM to reach ~US\$753bn by 2030e



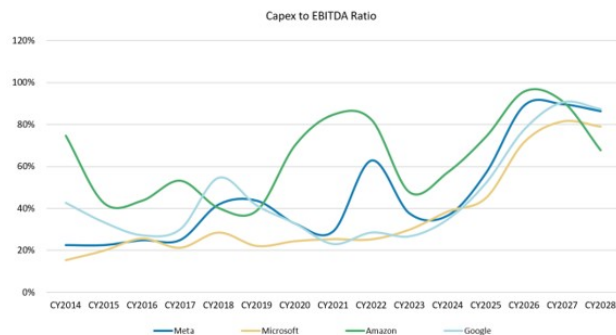
Source: Nvidia, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Cloud Semis: Brighter Outlook

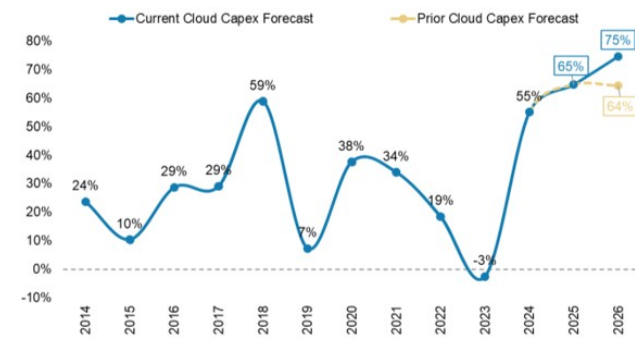
**Top 4 CSPs (Amazon, Google, Microsoft and Meta):
capex was up 95% Y/Y in 1Q26CY**



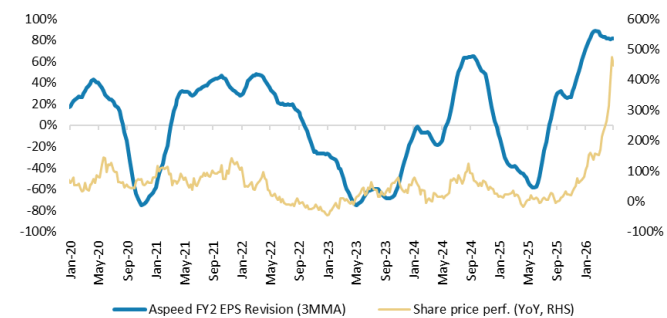
Capex-to-EBITDA ratio is reaching over 70%



Top 4 Cloud Providers: Cloud Capex Y/Y Growth



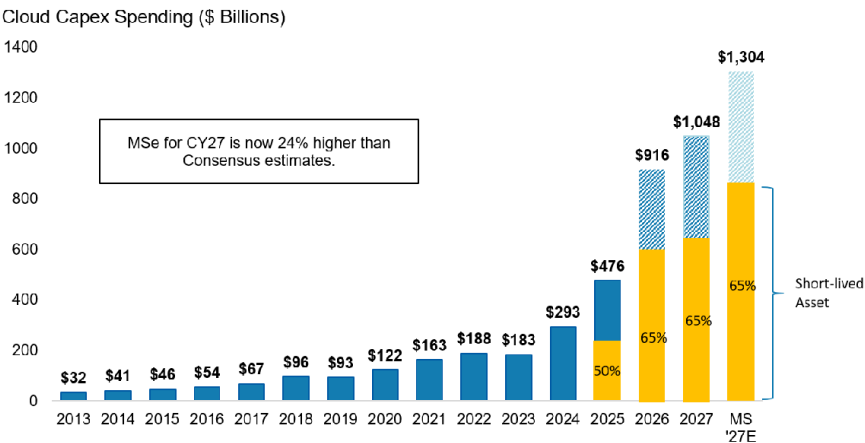
Aspeed – earnings estimate revision breadth



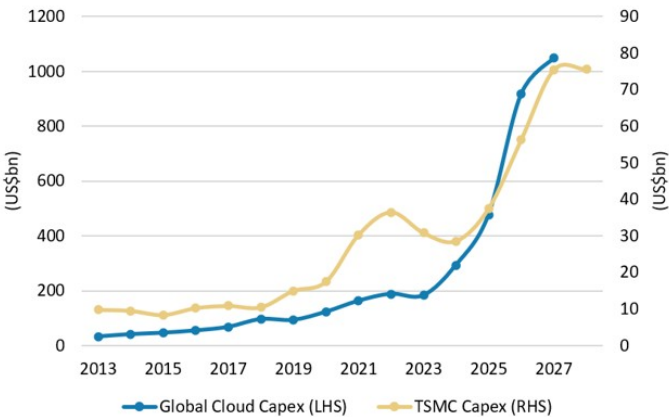
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Cloud Capex Remains Robust from Major CSPs

Morgan Stanley cloud capex tracker: we estimate nearly US\$1.3tn of cloud capex in 2027 (Top 14 listed global CSPs; no sovereign AI)



Global Cloud Capex vs. TSMC's Capex



Source: FactSet, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are top-down and from our US tech team.

Morgan Stanley

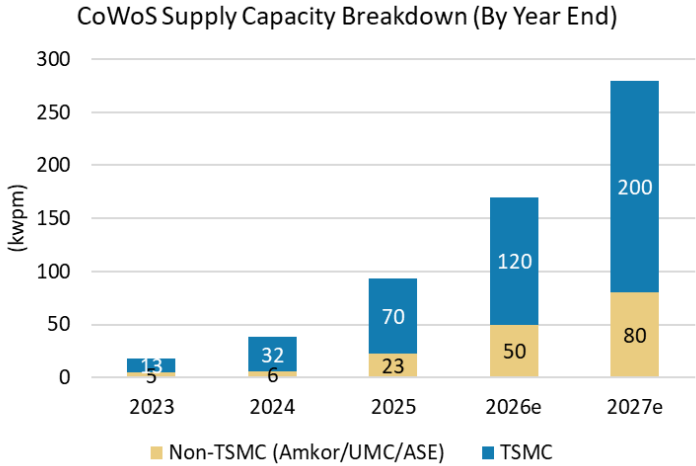
Announced Power Deployment – Implications for TSMC

GPU/ASIC Vendor	Announced Power Deployment Implication for TSMC			
	NVIDIA (Top US AI firm)	AMD (Top US AI firm)	Broadcom	AWS (Top US AI firm)
Power Deployed (GW)	10	6	3.5	2
Rack name	Vera Rubin NVL144	Helios	TPU	Trainium3 UltraServers
Power consumption per Rack (kW)	220	220	63	144
Rack Number (k units)	45	27	56	14
Chip Name	Rubin GPU	MI455 GPU	TPUv7 (Ironwood) TPUv8i (Sunfish)	Trainium3 (4)
Chip Volume (k units)	3,273	1,964	3,571	2,000
Life Cycle for the project	3	3	5	8
Implied life-cycle CoWoS Volume (k wafers)	409	166	260	118
Implied life-cycle 2/3nm wafer Volume (k wafers)	260	95	190	107
2027 annual CoWoS demand (k wafers)	136	55	52	15

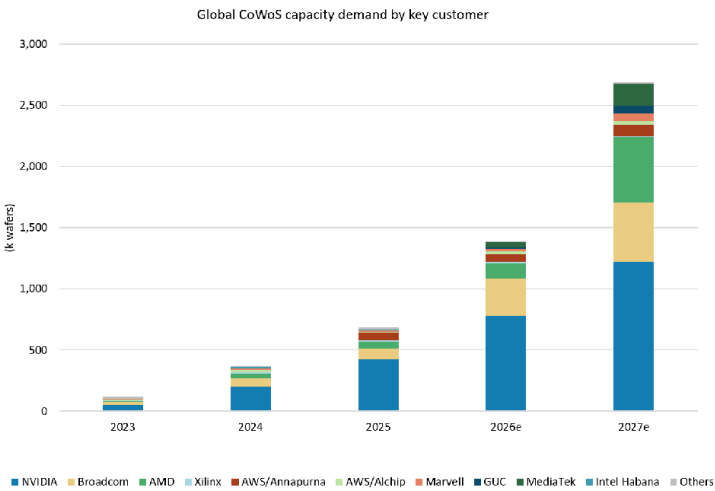
Source: Company data, Morgan Stanley Research; Note: Estimates are compiled using our Asia supply-chain checks. Note: We assume 64 units of TPU chips per rack with TPUv7 and v8i per chip TDP being ~980W.

TSMC May Expand CoWoS Capacity to 200kwpm by 2027 Given Ongoing Strong AI Demand

Global CoWoS capacity expansion by year-end and by vendor



Global CoWoS consumption, by customer



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Morgan Stanley

Global CoWoS demand breakdown

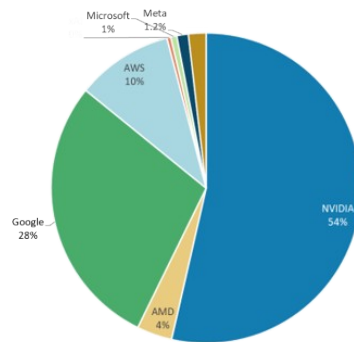
(k wafer)	2023	2024	2025	2026e	2027e	2023	2024	2025e	2026e	2027e
NVIDIA	53	200	425	780	1,222	45%	54%	62%	56%	45%
TSMC			390	680	1,090					
CoWoS-L			390	650	910					
CoWoS-S			0	20	50					
CoWoS-R			0	10	130					
Non-TSMC			35	100	132					
Amkor			35	100	132					
CoWoS-S			20	20	12					
CoWoS-R			15	80	120					
ASE/SPIL			0	0	0					
CoWoS-S			0	0	0					
CoWoS-R			0	0	0					
Broadcom	23	68	85	300	484	20%	18%	12%	22%	18%
TSMC			83	260	420					
CoWoS-L			0	15	55					
CoWoS-S			83	245	365					
ASE/SPIL			2	30	40					
CoWoS-S			2	30	40					
Amkor				10	24					
CoWoS-S				10	24					
AMD	7	40	60	130	530	6%	11%	9%	9%	20%
TSMC			60	80	240					
CoWoS-L			0	70	230					
CoWoS-S			60	10	10					
ASE/SPIL			0	50	170					
CoWoS-L			0	50	170					
Amkor				0	120					
CoWoS-L				0	120					
(k wafer)	2023	2024	2025	2026e	2027e	2023	2024	2025e	2026e	2027e
Xilinx	3	10	10	10	10	3%	3%	1%	1%	0%
MediaTek				40	180				3%	7%
TSMC				40	180					
CoWoS-S				40	180					
AWS/Annapurna			60	62	90				4%	3%
TSMC			60	32	54					
CoWoS-R			60	32	54					
ASE/SPIL				30	36					
CoWoS-R				30	36					
AWS/Alchip	9	16	5	26	36	8%	4%	1%	2%	1%
Intel Habana	0	7	9	0	0	0%	2%	1%	0%	0%
Marvell	1	18	15	17	64	1%	5%	2%	1%	2%
TSMC				5	50					
CoWoS-L				5	50					
CoWoS-R			15	0	0					
ASE/SPIL				12	14					
CoWoS-R				12	14					
GUC	1	1	2	14	60	1%	0%	0%	1%	2%
Cisco		2	3	5	6		1%	0%	0%	0%
Others	20	10	15	10	12	17%	3%	2%	1%	0%
Total demand	117	372	689	1,394	2,694	100%	100%	100%	100%	100%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

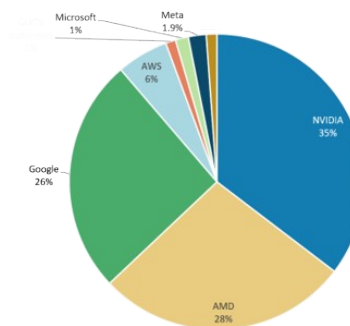
Morgan Stanley

AI Computing Wafer Consumption Could Reach >US\$46bn in 2027, with NVIDIA Accounting for the Majority

AI semi wafer consumption, by customer, 2026e



AI semi wafer consumption, by customer, 2027e

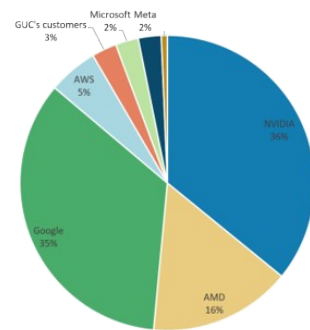


AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	Compute die size	Geometry	Compute die units	Wafer consumption (k wafers)	Wafer price (US\$)	Wafer revenue TAM (US\$ mn)
AI GPU (2027e)										
NVIDIA	B300	40	14	560	850	4nm	2	44	23,042	1,024
	Vera CPU	250	23	5,750	850	3nm	1	228	27,300	6,229
	Rubin R200	740	8	5,920	850	3nm	2	470	27,300	12,827
	Rubin Ultra	130	8	1,040	850	3nm	2	58	27,300	1,588
AMD	MI350 series	24	12	288	110	3nm	8	8	27,300	229
	MI455	157	7	1,099	850	2nm	2	13	30,000	400
	MI450 (for Meta)	35	14	490	850	2nm	1	3	30,000	89
	MI500 (Arcadia)	24	4	96		2nm			30,000	
	Venice CPU	270	21	5,670		2nm				
AI ASIC (2027e)										
Google	TPU v7p (Ironwood; AVGO)	13	16	208	700	3nm	2	14	27,300	371
	TPU v8i (Sunfish; AVGO)	330	12	3,960	800	3nm	2	296	27,300	8,075
	TPU v8t (Zebrafish; MediaTek)	180	20	3,600	800	3nm	2	269	27,300	7,341
	TPU v9 (Humufish; MediaTek)			400	850	2nm	2	32	27,300	867
AWS	Trainium 3	140	17	2,380	700	3nm	2	127	27,300	3,465
GUC's customers		60	20	1,200	645	4nm	1	29	23,042	674
Microsoft	Maia 300	50	11	550	850	2nm	1	29.1	30,000	873
Meta	MTIA 3 (Iris)	55	10	550	850	3nm	2	44	27,300	1,192
Others		18								
Total		2,664						1,700		46,208

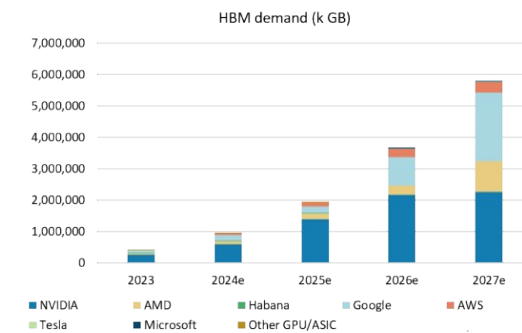
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are compiled using our supply-chain checks.

HBM Consumption in 2027e – Up to 48bn Gb

HBM consumption by customer in 2027e



NVIDIA still consumes most HBM supply in 2027e



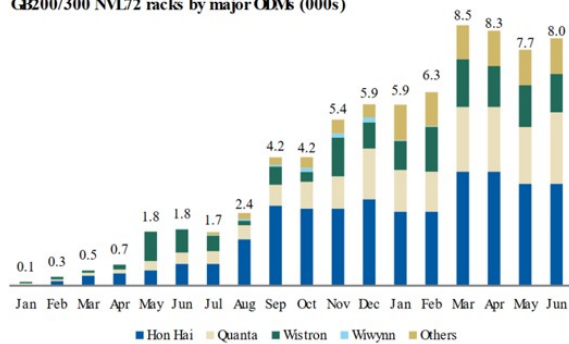
General assumptions				HBM assumptions						
AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	HBM chip density (GB)	HBM chip units	Total HBM size (GB)	HBM generation	HBM vendor	Total HBM demand (k GB)
AI GPU (2027e)										
NVIDIA	B300	40	14	560	36	8	288	HBM3e 12hi	Hynix/Micron	161,280
	Vera CPU	250	23	5,750	36	8	288	HBM4 12hi	Hynix/Micron/Samsung	1,704,960
	Rubin R200	740	8	5,920	36	8	288	HBM4e 12hi	Hynix/Micron/Samsung	399,360
	Rubin Ultra	130	8	1,040	48	8	384	HBM4e 12hi	Hynix/Micron/Samsung	82,944
AMD	MI350 series	24	12	288	36	8	288	HBM3e 12hi	Samsung/Micron	82,944
	MI455	157	7	1,099	36	12	432	HBM4 12hi	Samsung/Micron	474,768
	MI450 (for Meta)	35	14	490	36	6	216	HBM4 12hi	Samsung/Micron	105,840
	MI500 (Arcadia)	24	4	96	48	16	768	HBM4e 12hi	Samsung/Micron	73,728
	Venice CPU	270	21	5,670						
AI ASIC (2027e)										
Google	TPU v7p (Ironwood; AVGO)	13	16	208	24	8	192	HBM3e 8hi	Hynix/Samsung	39,936
	TPU v8i (Sunfish; AVGO)	330	12	3,960	36	8	288	HBM3e 12hi	Samsung/Hynix/Micron	1,140,480
	TPU v8t (Zebrafish; MediaTek)	180	20	3,600	36	6	216	HBM3e 12hi	Samsung/Hynix/Micron	777,600
	TPU v9 (Humafish; MediaTek)			400	48	12	576	HBM4e 12hi	Samsung/Hynix/Micron	230,400
AWS	Trainium 3	140	17	2,380	36	4	144	HBM3e 12hi	Hynix/Samsung/Micron	342,720
GUC's customers		60	20	1,200	24	6	144	HBM3e 8hi	Samsung?	172,800
Microsoft	Maia 300	50	11	550	36	8	288	HBM4 12hi	Samsung	158,400
Meta	MTIA 3 (Iris)	55	10	550	36	8	288	HBM3e 12hi	Samsung/Hynix/Micron	158,400
Others		18								
Total		2,664								6,077,256
Total HBM demand (mn Gb)										48,618

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

NVIDIA NVL72 Rack Forecasts

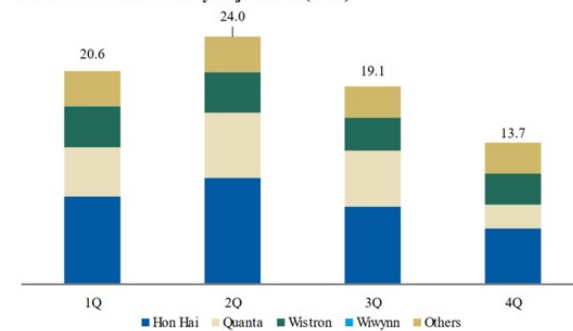
GB200/300 NVL72 Rack Ramp Monthly Trajectory

GB200/300 NVL72 racks by major ODMs (000s)

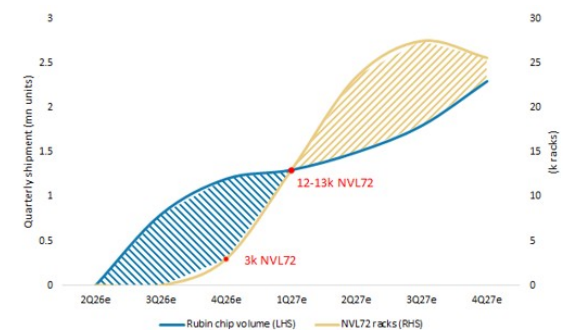
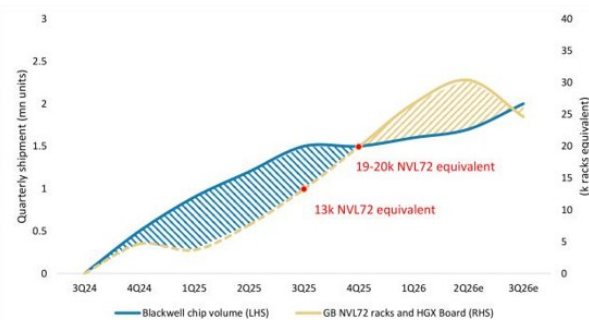


Industry-wide GB200/300 NVL72 Rack Split

GB200/300 NVL72 racks by major ODMs (000s)



We estimate that TSMC produced 5.1mn chips in 2025, with full-year GB200 NVL72 shipments seen reaching 30k

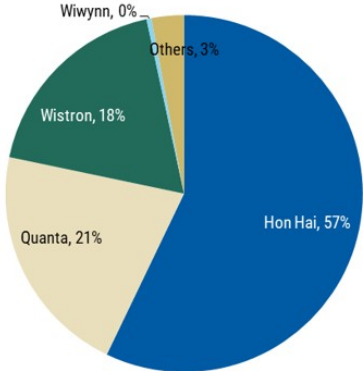


Source: Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: We think the actual rack deliveries to end-customers are lower than the numbers we have here because we include Wistron's computing tray (L10) rack equivalent number (without accounting for rack assembly and test times for L11). Estimates are from our Asia Pacific tech team's downstream estimates.

NVIDIA GB200/300 Server Racks – Supply/Demand Assumptions

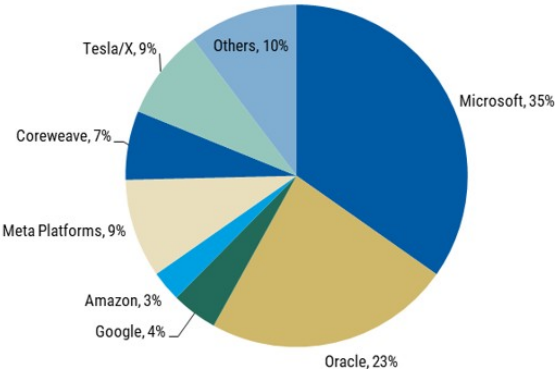
NVIDIA GB200/300 rack supply share (2025)

GB200/300 NVL72-equivalent rack supply share



NVIDIA GB200/300 rack demand share (2025)

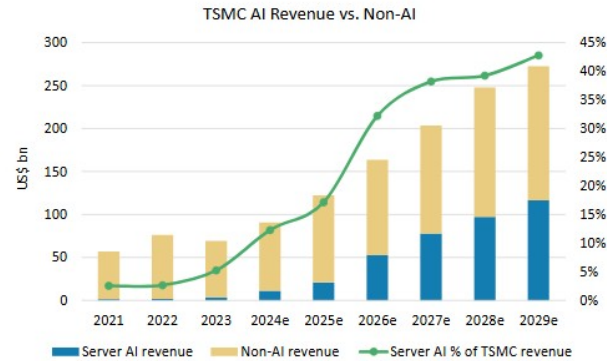
Nvidia GB200 server demand share, 2025e



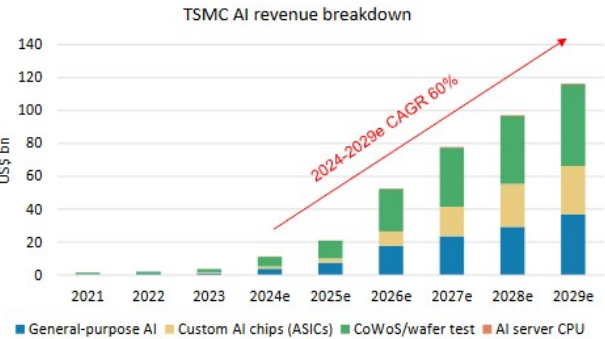
Source: Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are from our Asia Pacific tech team's downstream estimates.

TSMC – AI Semis Revenue Could Reach 60% from 2024 to 2029e

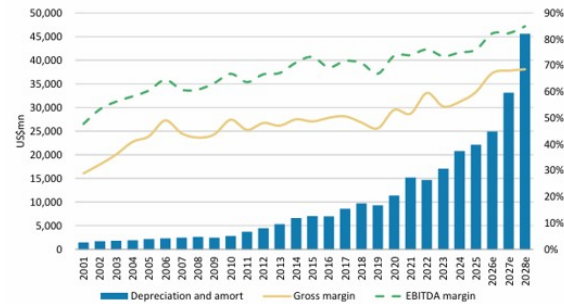
TSMC – AI semis revenue could account for >30% of 2026e revenue



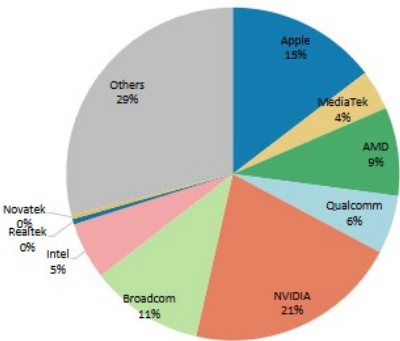
TSMC – AI semis revenue breakdown



TSMC – margin expansion



TSMC – 2026e revenue breakdown by customer

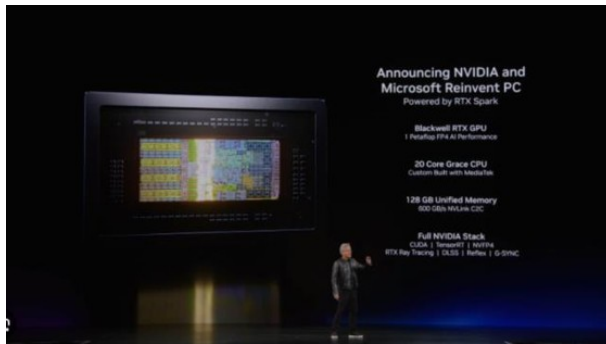


Source: Company data, Morgan Stanley Research (e) estimates

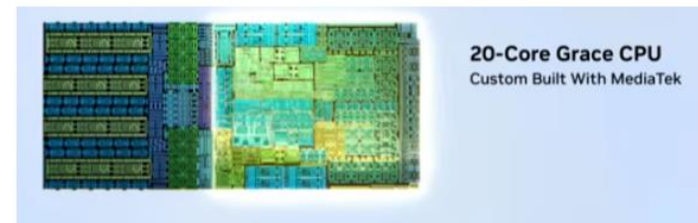
Morgan Stanley

Nvidia/MediaTek – New AI PC (RTX Spark) Could Trigger Potential Edge Device Replacement

New AI PC announced by Nvidia (N1X)



MediaTek provides 20-core customized Grace CPU for N1X



RTX Spark (N1X) provided by MSI (available in 3Q26)



RTX Spark (N1X) provided by ASUS (available in 3Q26)

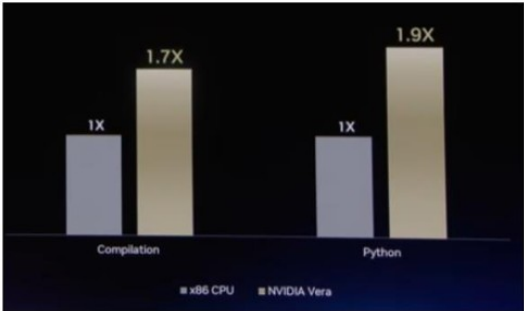


Source: Nvidia, MSI, ASUS

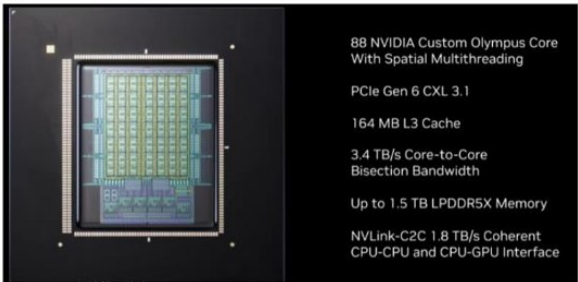
25

Nvidia’s Latest Vera CPU Could Support Robust Agentic CPU Demand

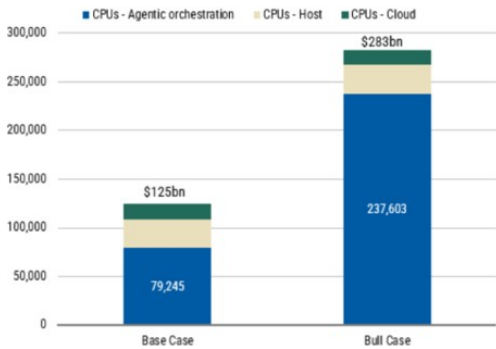
Vera performance could be 1.8x that of highest-performance x86 CPU



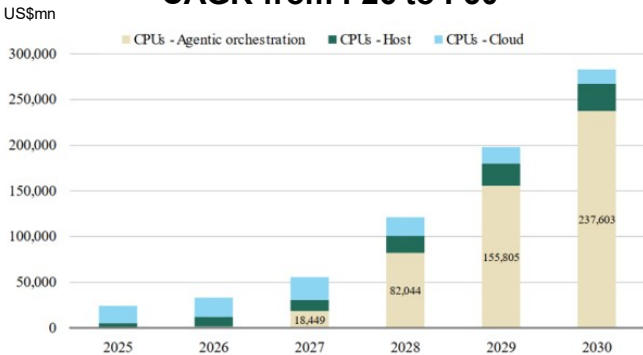
Vera CPU cores are not split across chiplets, enabling faster core-to-core connection



Our bull case implies a US\$238bn CPU orchestration TAM



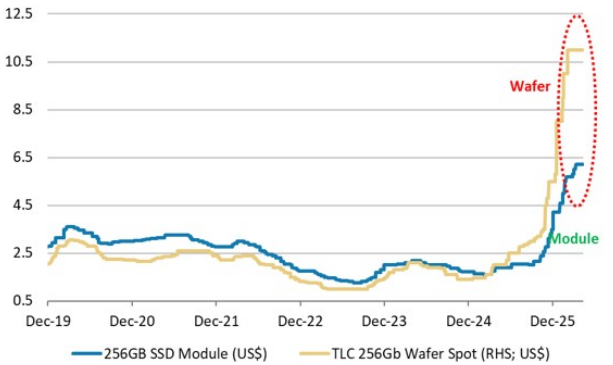
In MS top-down model, we estimate the Agentic CPU TAM will expand at a 251% CAGR from F26 to F30



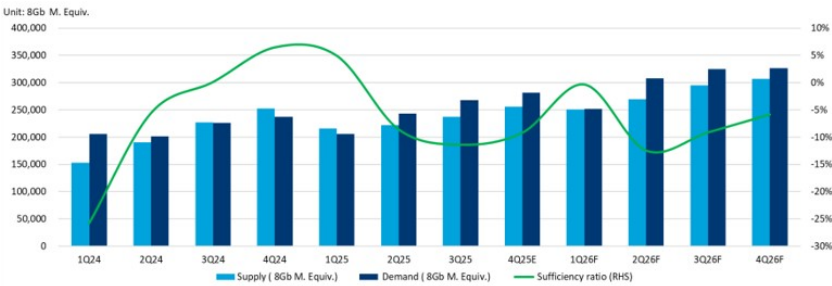
Source: Nvidia, Morgan Stanley Research (e) estimates from our Global team; see [Global Technology: Agentic AI – The Surge Begins \(11 May 2026\)](#)

AI Storage Leads to NAND Shortage; We Also Expect NOR Flash Undersupply into 2026

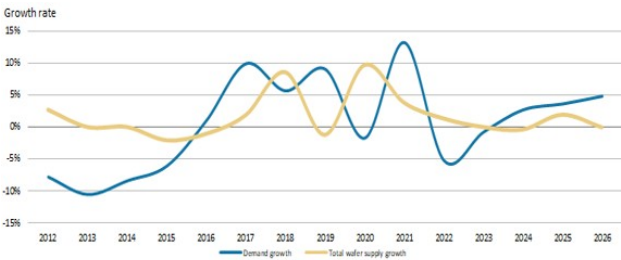
NAND wafer spot pricing and module prices



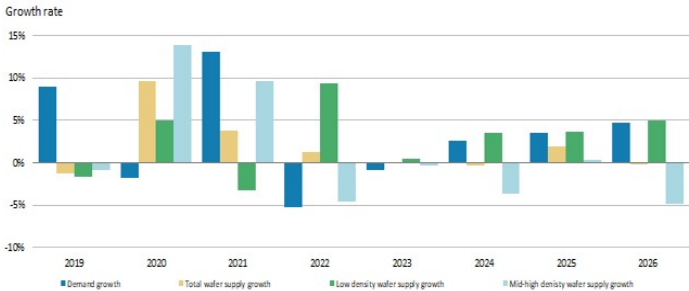
NAND supply and demand model



NOR flash demand and supply growth rates



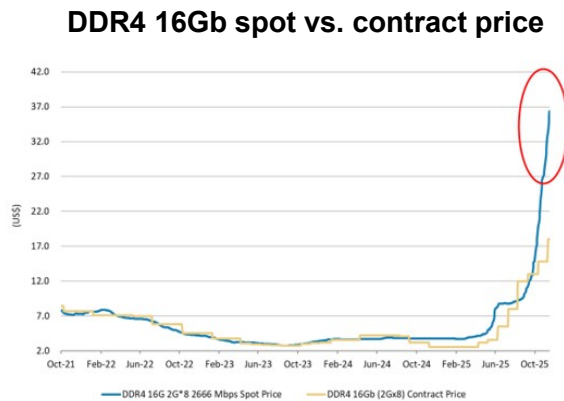
NOR flash demand and supply growth rates



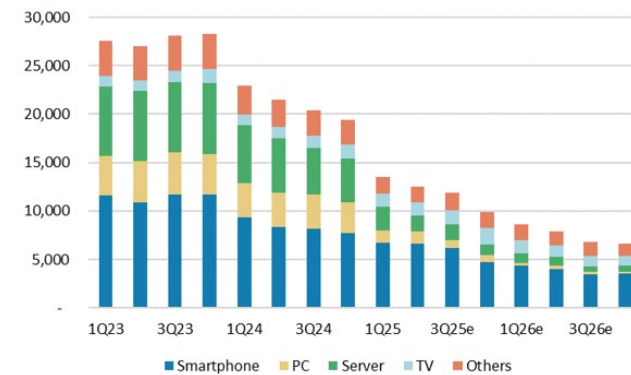
Source: ChinaFlashMarket, Morgan Stanley Research estimates.

Morgan Stanley

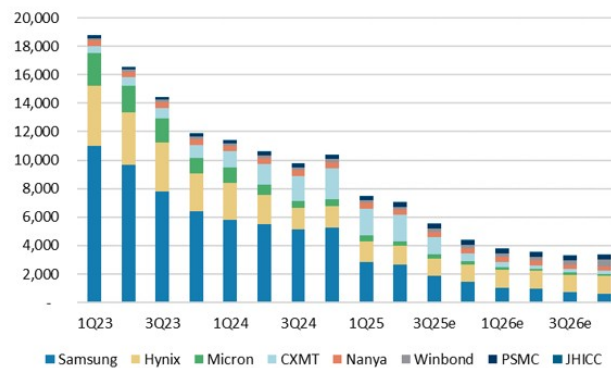
DDR4 Shortage to Persist into 2H26; Spot Price Capped



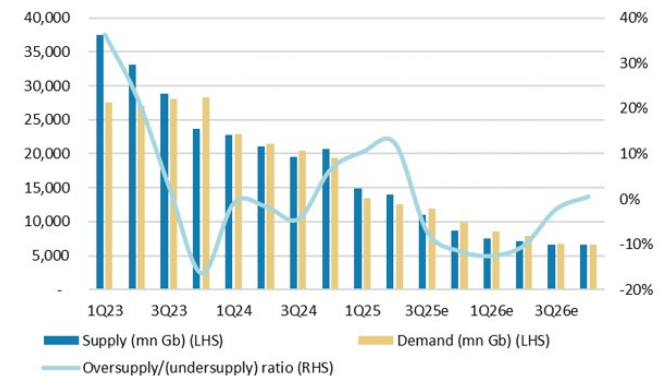
Quarterly demand breakdown, by product (mn Gb)



Quarterly supply breakdown (mn Gb)



Quarterly supply and demand summary



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Morgan Stanley

AI ASIC, CPO, and Chip Testing

Morgan Stanley

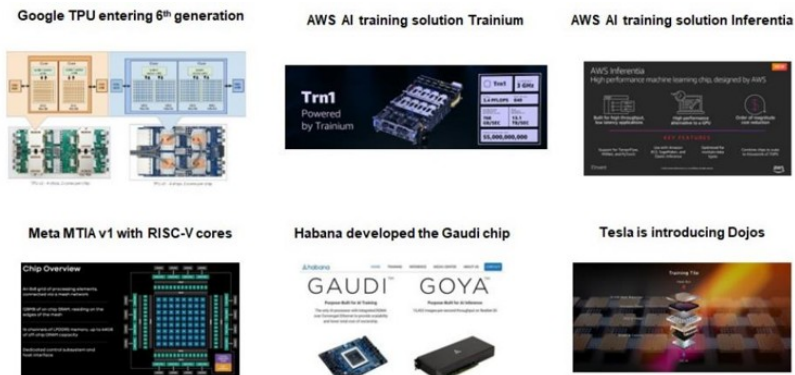
AI Semis: Now and the Future – “Prompts”

- Generative AI Demand Driver:
 - Killer apps
 - Competition
 - Sovereign
- Growth limitations:
 1. Budget
 2. Energy -> in the US
 3. Chip Capacity -> in China
 4. Regulation
- Semiconductor solutions:
 - Moore's Law
 - CoWoS/ SoIC
 - HBM
 - CPO
 - Custom chip
 - GaN HVDC 800V
- Growth perspective:
 1. Inference vs. Training AI semis
 2. Edge vs. Cloud AI semis
 3. Custom ASIC vs. AI GPU

Morgan Stanley

CSPs Still Need Custom Chips, Even with NVIDIA Providing Powerful AI GPUs

Global CSP cloud AI custom chips



The volume upside of AI chips like Google TPU could depend on ABF substrate supply



Trainium Forecasts from Our Asian Research team

k Units	2023	2024	2025e	2026e	2027e	2028e
Trainium1/Inferentia2	300	600	30			
Trainium2		300	1,200			
Trainium3			85	1,700	2,380	-
Trainium4					150	2,000
Total	300	900	1,315	1,700	2,530	2,000

Source: AWS, Google, Meta, Intel, Tesla, Morgan Stanley Research (e) estimates.

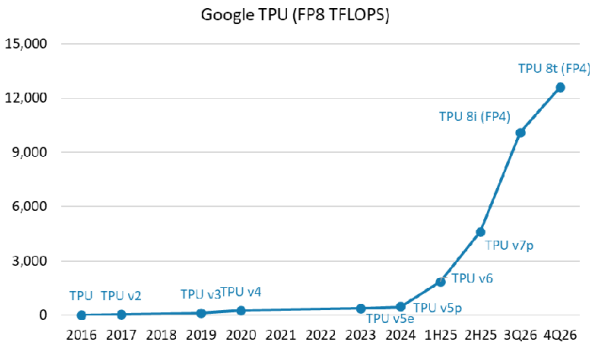
31

More ASIC Projects Are Coming, According to Each CSP Plan

TPU Forecasts from MS Asia Research team

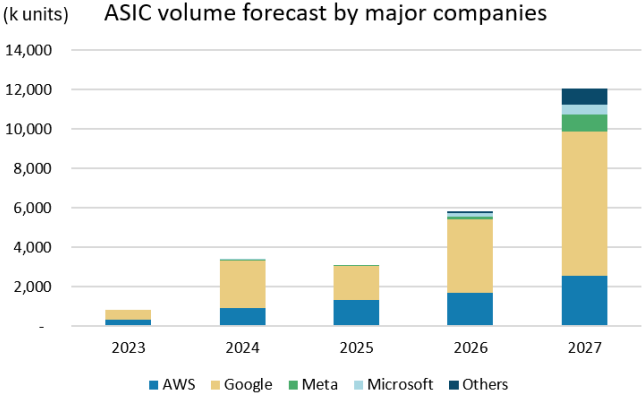
k Units	2023	2024	2025e	2026e	2027e	2028e
v5	500	2,400	250			
v6 (Trillium)			1,200			
v7 (Ironwood, by Broadcom)			300	2,300	200	
v8i (Sunfish; 3nm, by Broadcom)				900	4,000	2,500
v8t (Zebrafish; 3nm, by MediaTek)				500	3,000	1,000
v9 (Humufish; 2nm, by MediaTek)					150	3,000
v9a (Merope; 2nm, by US design service)						unknown
v10 (Icefish; 1.4nm, by MediaTek)						unknown
Total	500	2,400	1,750	3,700	7,350	>6500

Google's TPU performance is growing rapidly



Source: Company data, Morgan Stanley Research (e) estimates

ASIC volume forecast by major companies



Morgan Stanley

KYEC benefits from strong AI GPU, TPU and CPU growth

AI Revenue Forecast For KYEC

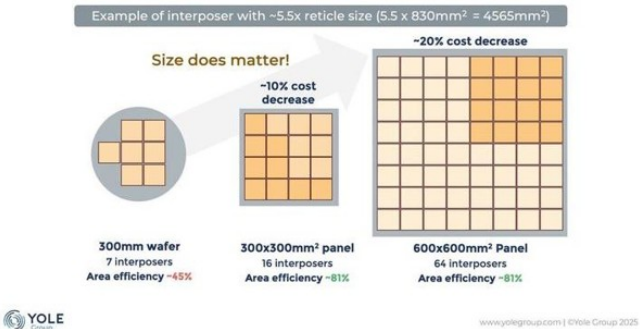
	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2025	2026e	2027e
Units (k)											
H-series	650	200	250	500	400	400	600	600	1,600	2,000	
B-series with GDDR7	-	-	300	500	250	250	250	250	800	1,000	
B-series with HBM	1,000	1,100	1,400	1,500	1,500	1,500	1,600	300	5,000	4,900	560
R-series							800	1,200		2,000	6,500
G-series CPU									1,100	2,450	250
V-series CPU										2,500	5,500
Google TPU									1,750	3,680	6,000
Google CPU										2,000	2,500
Test time											
Per chip test time (seconds)											
H-series test time	350	350	350	350	350	350	350	350	350	350	350
B-series with GDDR7 test time			350	350	350	350	350	350	350	350	350
B-series with HBM test time	700	750	900	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
R-series test time				1,200	1,200	1,200	1,200	1,200		1,200	1,800
G-series CPU test time									300	300	300
V-series CPU test time										700	700
Google TPU blended test time									186	340	450
Google CPU test time										300	300
Total test demand (k hrs)											
H-series	63	19	24	49	39	39	58	58	156	194	-
B-series with GDDR7			29	49	24	24	24	24	78	97	-
B-series with HBM	194	229	350	417	417	417	444	83	1,190	1,361	156
R-series					-	-	267	400		667	3,250
G-series CPU									92	204	21
V-series CPU										486	1,069
Google TPU									90	348	750
Google CPU										167	208
Test hour rate (for the largest customer including burn-in)	170	170	170	170	170	170	180	190	170	178	200
Test hour rate (for TPU with self-bought testers)									200	250	280
Largest customer's AI revenue to KYEC (US\$ mn)	43.8	42.3	68.6	87.4	81.6	81.6	142.9	107.5	257.6	536.1	899.2
% of KYEC MSe Revenue (quarterly figures without CPU)	19%	15%	23%	27%	25%	22%	34%	23%	23%	34%	40%
TPU revenue to KYEC (US\$ mn)									18.1	86.9	210.0
% of KYEC MSe Revenue									1.6%	5.5%	9.3%
Google CPU revenue to KYEC (US\$ mn)										41.7	58.3
% of KYEC MSe Revenue										2.6%	2.6%

Source: Company data, Morgan Stanley Research (e) estimates

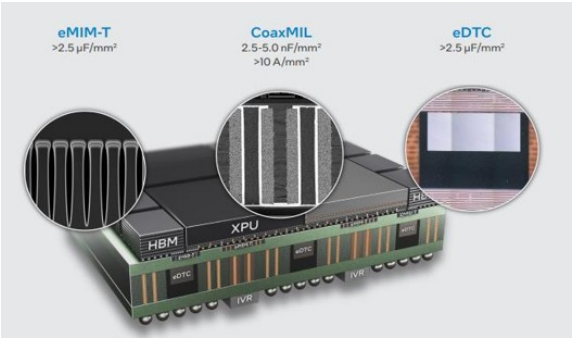
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What About the Competition Between TSMC’s CoWoS and Intel’s EMIB?

TSMC CoWoS can support up to 9.5x
reticles, or four chips per wafer

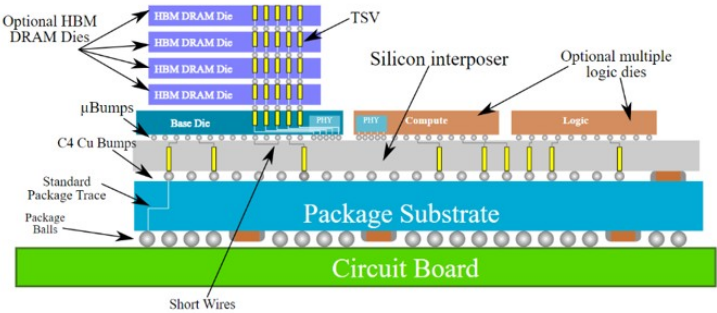


Intel EMIB eDTC: APMemory’s technology

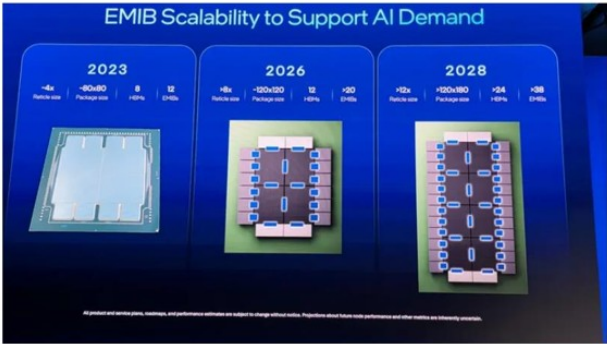


Source: Intel, Yole, Morgan Stanley Research

TSMC’s CoWoS-S (silicon interposer)
packaging architecture

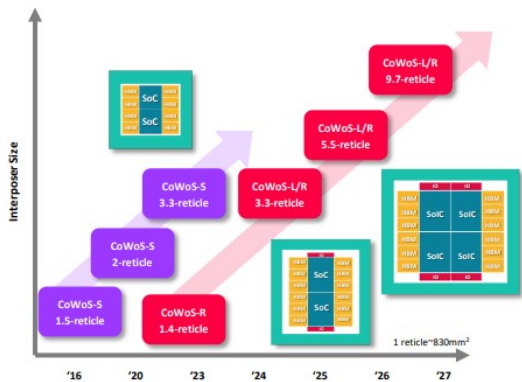


Intel’s EMIB can easily support larger chips with more
reticles (>12) if its supply chain can execute well

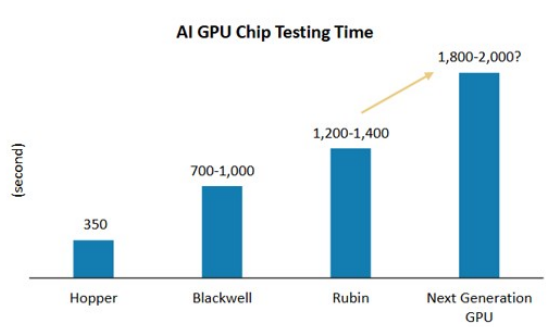


Larger Package Sizes Are Emerging as a Key Industry Trend

TSMC's roadmap for interposer size



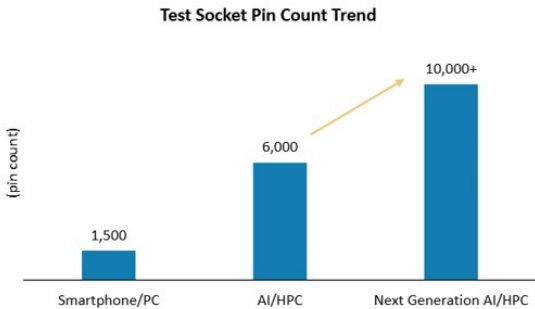
Chip testing time continuing to trend up



Testing equipment market is expected to expand at a 35% CAGR during 2024-27e

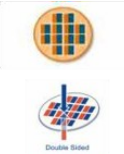
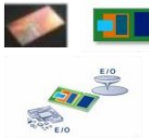
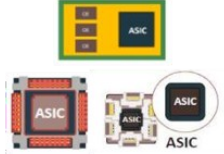


AI/HPC test socket needs higher pin counts vs. traditional consumer tech applications



Source: Company data, Morgan Stanley Research (e) estimates

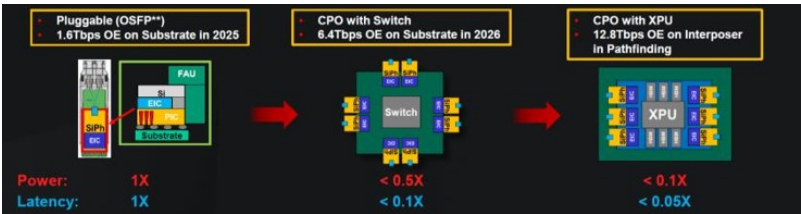
New Evolution for Testing Equipment and Components: CPO

Test Insertion	Insertion 1	Insertion 2	Insertion 3	Insertion 4		Insertion 5 (potential)
				Insertion 4O	Insertion 4E	
					CPO package (ASIC+OE)	
Test content	Electronic IC (EIC) & Photonic IC (PIC)	EIC die to PIC wafer (after SoIC)	Optical engine (die-to-die)	CPO package (ASIC+OE) Optical Testing	Electrical Testing (2025-2026) Optical+Electrical Testing (starting 2027e)	CPO package (ASIC+OE) SLT?
Testing surface						
Key Product timeline	2025-2026 Scale-out CPO switch (Spectrum+Quantum) products 2027e onward: Scale-out CPO switch (Spectrum+Quantum) + Scale-up CPO switch products					
Testing details	El. & opt. DC, (power, loss, dark current etc.)	E/O, O/E, O/O, high speed, S-parameters	Full calibration/DC, high-speed functional, optical loopback, alternatively: S-parameters	Optical Light Transmission Testing, optical loopback	BER testing, signal testing	Full system functional validation
Testing service provider	Foundry (Wafer-level) TSMC	Foundry (Wafer-level) TSMC	OSAT (Die/chip level) SPIL (ASE Group)	OSAT (Die/package level) SPIL (ASE Group)	OSAT (Die/package level) SPIL (ASE Group)	OSAT (Die/package level) SPIL (ASE Group)
Equipment and Consumable Vendor	PIC ATE tester: 1) Advantest + TEL + Formfactor 2) Chroma (underqualification) EIC tester: 1) Teradyne + TEL EIC Probe card: FormFactor	Wafer level testing: 1)Ficontec +Formfactor +Teradyne 2) Advantest+ MPI	Die-to-die prober: 1. TEL 2. MPI Optical engine E/O tester+laser reliability test+ELS source test head: Chroma Optical and Electrical Test Socket: Winway	CPO switch tester: Chroma Coaxial socket/Hypersocket: Winway	ATE tester: 1) Advantest (under qualification) 2)Teradyne FT Handler+ELS source test head+optical alignmnet stage : Hon Precision Coaxial socket/Hypersocket: Winway	CPO SLT tester: Chroma (under qualification) Coaxial socket/Hypersocket: Winway

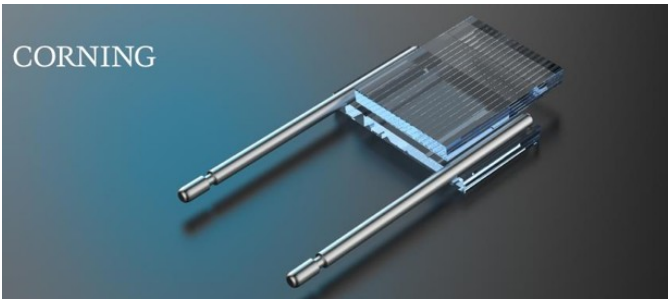
Source : Company data, Morgan Stanley Research

CPO Supply Chain Updates; More on GlassBridge

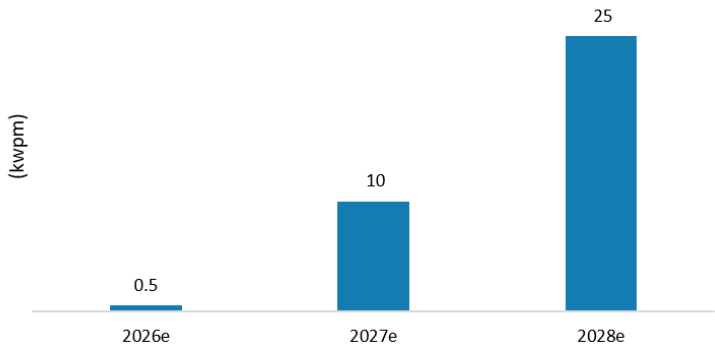
TSMC's COUPE technology



GlassBridge Fiber-to-PIC Connector



TSMC PIC capacity planning



PIC Implied Optical Engine shipment

	2026e	2027e	2028e
TSMC PIC capacity (kwpm)	0.5	10	25
Die per wafer	648	648	648
Annual PIC production (mn)	4	78	194
SoIC yield rate	50%	50%	50%
Optical engine production (mn)	2	39	97
Downstream assembly yield rate	20%	20%	50%
Actual Optical engine shipment (mn)	0.39	7.78	48.60

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Morgan Stanley

CPO Supply Chain Updates; More on GlassBridge

Upcoming key CPO projects FAU vendors

CPO project	FAU Vendor	Introduced Timeline
NVIDIA Scale-out (Spectrum/Quantum)	TFC, FOCI, Senko?	2H26
NVIDIA Scale-up (OIO)	FOCI, Senko	2029 (Feynman Ultra)
Broadcom scale-out (Tomahawk)	Twinstar, Senko	2H24
AMD scale-up (MI series)	FOCI, Senko, Teramont?	2H27-2028 (MI500 small scale; MI600 mainstream)
Ayar Labs	FOCI, Senko	2H27-2028 (standard solution); AWS Trainium 5 (2H28-2029)
Marvell (COUPE version)	FOCI, Senko	2028
MediaTek		2028?

Bottom-up analysis: NVIDIA's revenue contribution to FOCI in 2026-28e

	2026e	2027e	2028e
Rubin Ultra rack unit (k units)		5	20
Optical Engine content per Rubin Ultra rack		792	792
Scale-Up (NVLink switch/Quantum)		648	648
Scale-out (CX10)		144	144

Other scale-out switch optical engine consumption

Spectrum-X (k units)	800	1,000	2,112
Quantum-X (k units)		1,000	1,408

Market Share Assumptions

FOCI	35%	35%	35%
Others (TFC, Senko)	65%	65%	65%

FAU ASP Assumptions

FAU ASP (6.4T; US\$)	180	180	180
FAU ASP (3.2T; US\$)	90	90	90
FAU ASP (1.6T; US\$)	45	45	45

FOCI Revenue contribution from NVIDIA

Revenue contribution (US\$mn)	19	219	721
Revenue contribution (NT\$mn)	567	6,577	21,622
FOCI CY Revenue (NT\$mn)	1,985	8,694	23,559
% of total revenue	29%	76%	92%

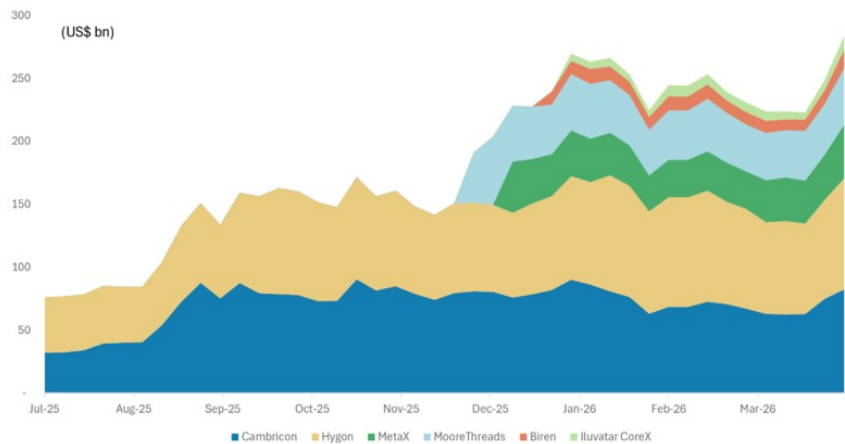
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Morgan Stanley

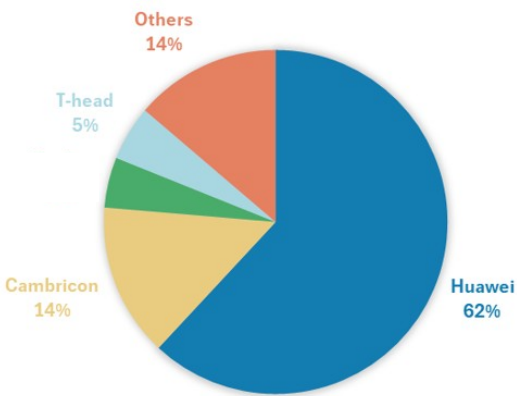
China – AI Computing

Growing Market Size and Share of Domestic AI Semis

China’s AI GPU market cap is growing, with more IPOs to come



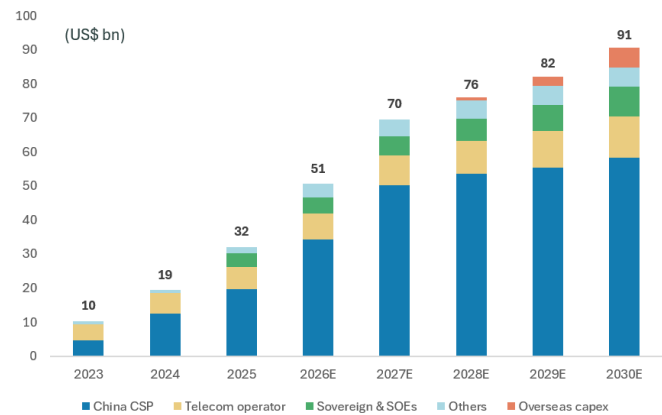
Domestic AI accelerator market share in 2026e



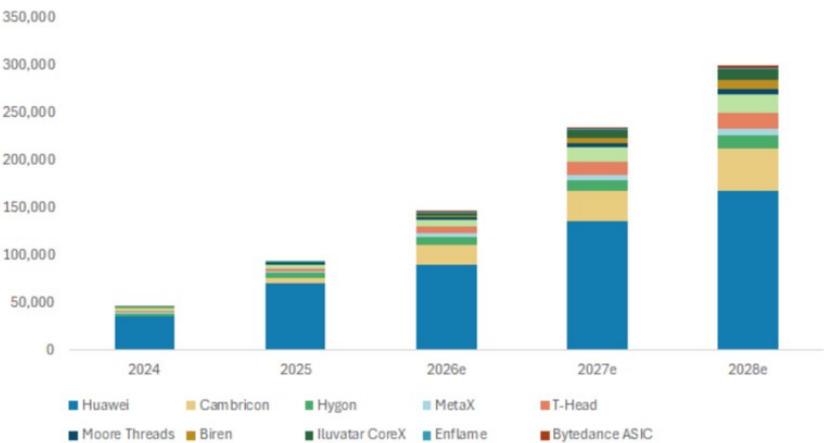
Source: Bloomberg, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

We Expect China’s AI GPU TAM to Grow to US\$91bn by 2030e

We expect China’s AI chip TAM to grow to US\$91bn by 2030



AI accelerators' revenue trend



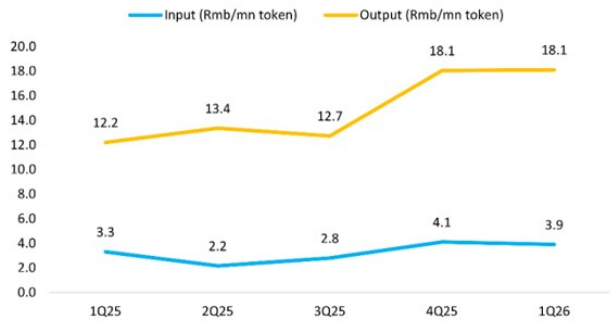
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Near-term market tracker for China’s AI GPU demand

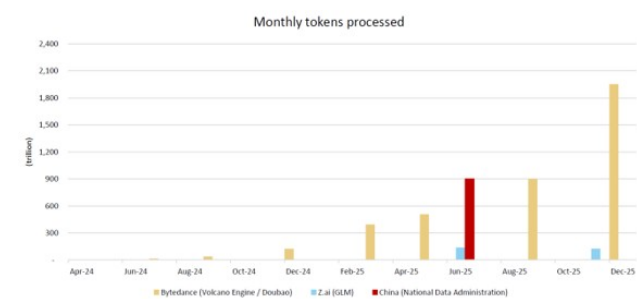
Nvidia's 5090 price keeps rising in China



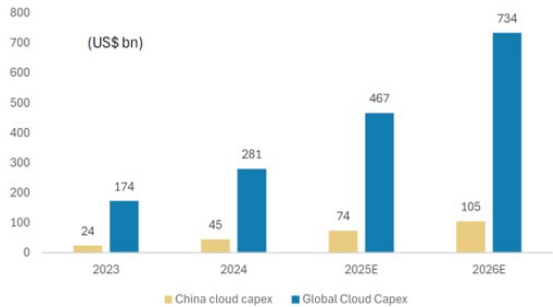
Average token price for China's mainstream AI LLMs



Surge in ByteDance (Volcano Engine/Doubao) tokens indicates high AI demand



China vs. global – cloud capex trend



Source: Company data, Taobao, Morgan Stanley Research.

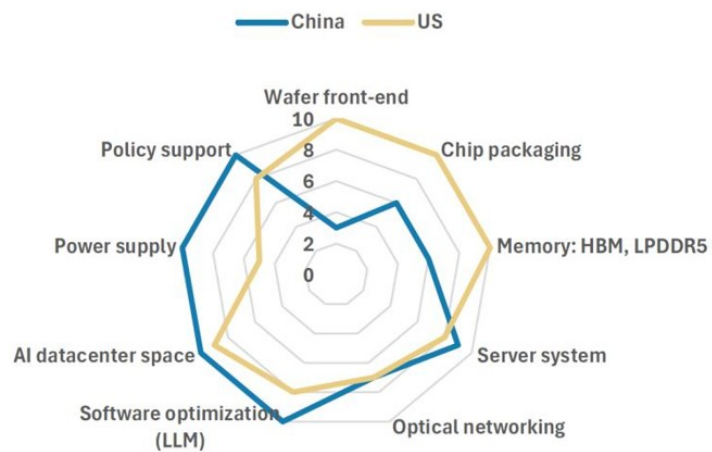
AI Chip Value Chain – China vs. the US – A Decoupling of AI Computing



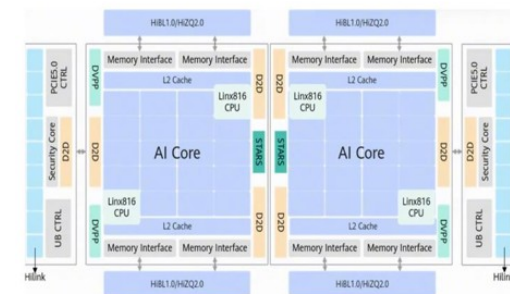
Source: Morgan Stanley Research.

China's Infrastructure Strength Narrows Perceived Technology Gap

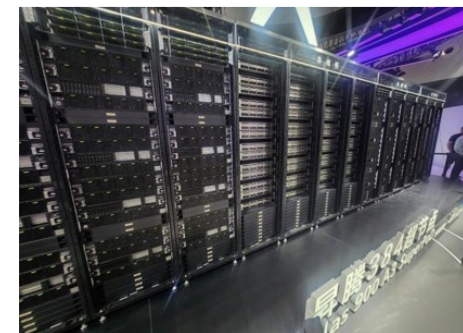
Nine factors comparison between US and China AI – Chip, System, Infrastructure



Die shot of Ascend 950



Huawei CloudMatrix 384 A3 SuperPod



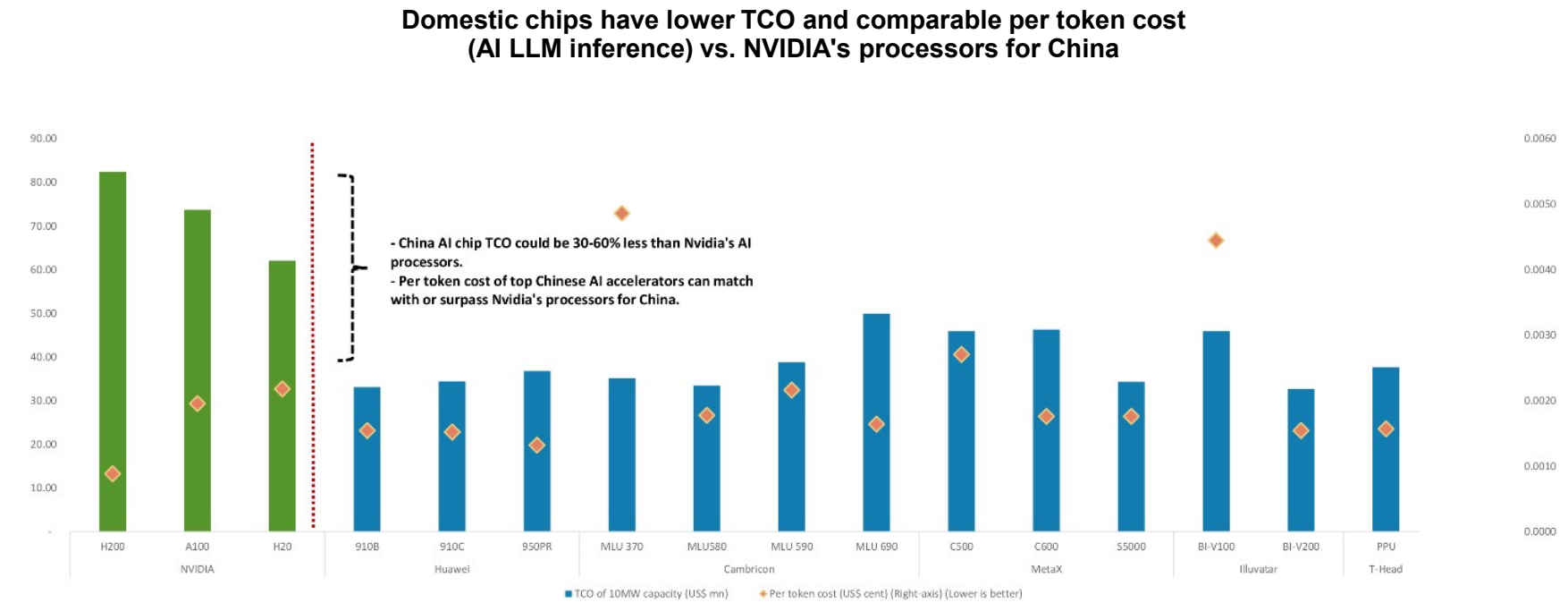
Step 1: If one computing die is not powerful, package more dies into a single chip.

Step 2: If one chip is not powerful enough, build larger racks and clusters.

Step 3: If one fab is not sufficient, expand manufacturing capacity.

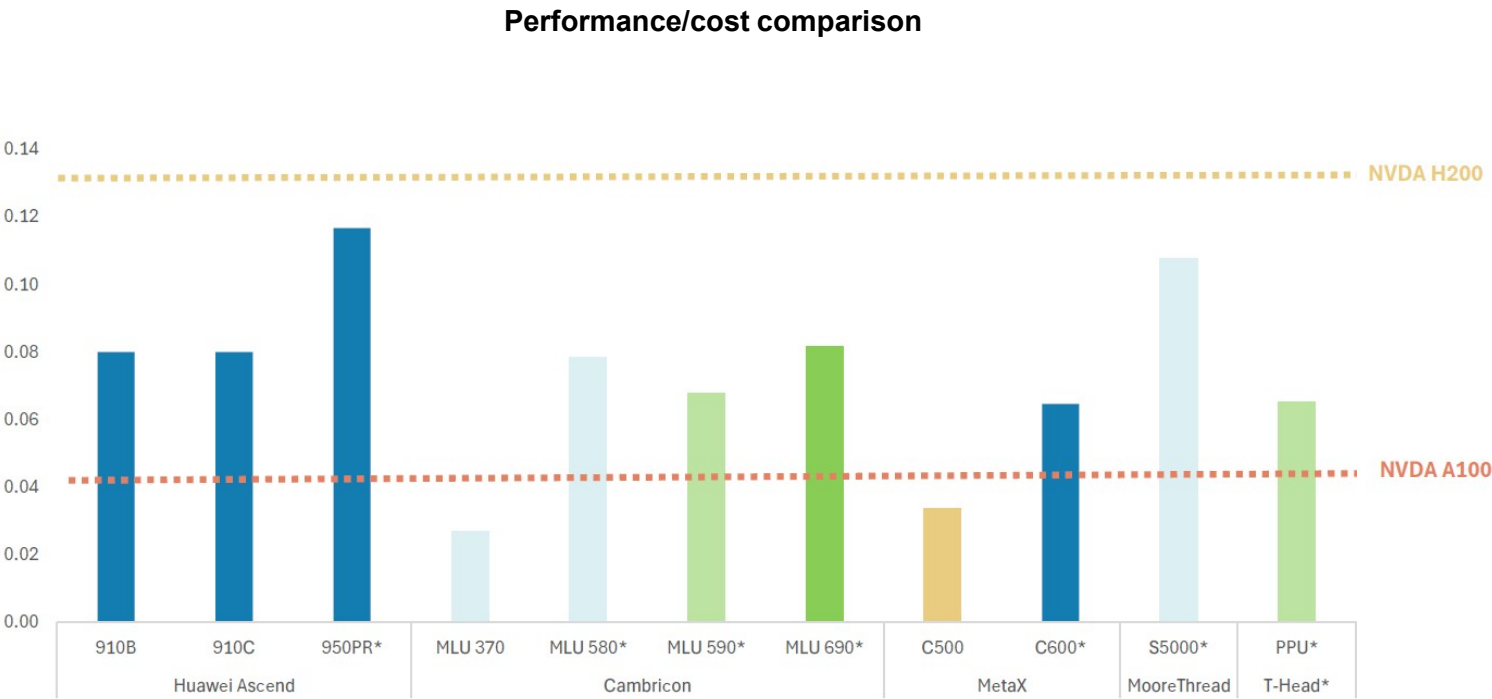
Source: Morgan Stanley Research.

Inference Economics: TCO and Cost per Token



Source: Company data, Morgan Stanley Research estimates.

Domestic Chips Deliver Stronger Performance per Dollar Thanks to Materially Lower Pricing



* = Based on Morgan Stanley estimates. Source : Company data, Morgan Stanley Research.

Morgan Stanley

Huawei Ascend Atlas 950: scaling the SuperPod beyond 1,000 NPUs

Huawei showcased its new Atlas 950 SuperPod at 2026 WAIC



NPU blade (tray) of Atlas 950 SuperPod



The reverse side of Atlas 950 mainly showcases the optical interconnects adopted by UBLINK



Source: Company data, Morgan Stanley Research.

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A flourishing SuperPod landscape at WAIC 2026

Comparison of some key SuperPod solutions at 2026 WAIC

Solution	Vendor / Partner	Scale	Architecture Highlights	Key Differentiator
Atlas 950	Huawei	1,024 Ascend 950DT NPU	16 compute cabinets + 4 UnifiedBus cabinets; UnifiedBus 2.0 unified memory semantics; hybrid copper + optical interconnect; up to 256TB pooled memory	Largest scale-up domain showcased at WAIC, emphasizing large-model training and unified memory architecture
MTT C256 SuperPod	Moore Threads	128-256 S series GPGPU	Single-level scale up with cabletree connections.	Focus on scalable domestic GPU clusters for AI training and inference
Yunsui SL64-O	Enflame + ZTE	64 L600 AI accelerators	Co-developed rack-level solution using midplane-free orthogonal architecture integrating compute, networking and cooling	Demonstrates ODM-driven ecosystem collaboration and modular SuperPod deployment
Xijing S6000	MetaX + ZTE	64 C600 AI accelerators	Similar as Yunsui SL64-o, using midplane-free orthogonal architecture	Equipped with latest MetaX AI training + inference accelerator, C600 chip

Midplane-free orthogonal designs at WAIC



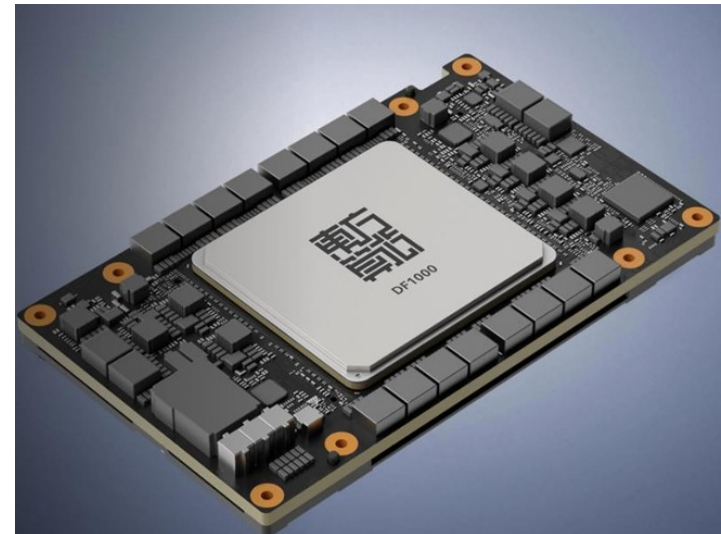
Source: Company data, Morgan Stanley Research.

New AI accelerators introduced at WAIC 2026

Iluvatar unveiled Tiangai 300 for full-stage AI inference



Oriental Computing: 3D stacking offers an alternative path beyond process scaling



Source: Company data, Morgan Stanley Research.

Morgan Stanley

Comparison among Cambricon, MetaX, and Iluvatar

			
Ticker	688256-SH	688802-SH	9903-HK
Product	MLU 220/270/370/580/590/690 (AI Training + inference)	C Series (AI training + inference) N Series (AI Inference) G Series (Graphic rendering)	Tiangai 100/150/200/300 (AI training + inference) Zhikai 100 (AI inference)
GPGPU/ASIC	ASIC	GPGPU	GPGPU
Chip suppliers			
Process node for latest products	7nm/N+2	12nm/N+1	7nm
Secured orders from major CSPs	✓	✗	✓
Sovereign fund as major shareholder	✗	✓	✗
Per token cost performance	↑	→	→
2025 Revenue (Rmb mn)	CNY 6,497	CNY 1,644	CNY 1,034
Profitability	✓	✗	✗

Source: Company data, Morgan Stanley Research.

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(as of June 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$88.42
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb375.20
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$123.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,510.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$649.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,249.00
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$4,380.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,120.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$480.50
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$149.40
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$544.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$286.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb72.83
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,875.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb813.78
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$426.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb740.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb96.66
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$1,920.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb103.86
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$474.50
SMIC (0981.HK)	O (10/21/2025)	HK\$70.75
TSMC (2330.TW)	O (02/07/2022)	NT\$2,405.00
UMC (2303.TW)	O (05/19/2026)	NT\$138.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$168.50
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$364.50

Daisy Dai, CFA

ASMPT Ltd (0522.HK)	O (07/24/2025)	HK\$166.50
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb61.12
Emphyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb93.06
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb32.22
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb312.00
Innoscence (2577.HK)	E (10/13/2025)	HK\$45.78
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb80.81
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$26.16
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb97.32
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb84.84
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb65.20
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb24.48
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb93.14
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$904.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,405.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$15,540.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$99.70
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$171.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb103.92
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb451.96
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$129.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$311.80
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb225.50
Novatek (3034.TW)	U (02/04/2026)	NT\$503.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$141.00
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$612.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$66.00
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$745.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb49.72
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$161.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$121.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$207.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb113.90
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb375.85
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,025.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$589.00

Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$13.27
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,465.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$6,250.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$289.08
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,610.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.